

Sustainability Aware Asset Management

Asset Allocation with a Carbon Objective

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Abstract

Can deep portfolio decarbonisation be achieved without eroding financial performance? Using monthly returns and annual CO₂ data for more than 600 North-American equities (2014–2024), we extend mean–variance optimisation into a three-dimensional (μ, σ , carbon) space and evaluate five allocations: (i) an unconstrained minimum-variance portfolio; (ii) a value-weighted benchmark; (iii) a minimum-variance design capped at 50 % of its own footprint; (iv) a benchmark-neutral strategy that halves the benchmark footprint subject to a tracking-error budget; and (v) a dynamic Net-Zero portfolio imposing a 10 % yearly emissions decline. Across the sample, carbon-constrained strategies cut financed emissions by 20–50 % while delivering Sharpe ratios statistically indistinguishable from—sometimes superior to—the unconstrained benchmark. Their annualised tracking errors cluster around 4.9 %, well above the 0.30–0.50 % floor to prevent index replication. Only the Net-Zero design at the shortest look-back window (48 months) briefly peaks at 7.4 % before reverting below 5 % as the estimation horizon lengthens. Early cuts of up to roughly 50 % in carbon footprint appear to entail little to no performance penalty, whereas deeper reductions trade off transparently against the tracking-error “budget”. The Net-Zero trajectory remains consistent with a 1.5 °C carbon budget while preserving almost the entire equity risk premium, offering counter-evidence to the notion of an inevitable sustainability tax. Overall, the evidence shows that ambitious, Paris-aligned decarbonisation can be embedded in mainstream portfolios through explicit tracking-error limits, offering investors a coherent and auditable framework for aligning assets with the Paris Agreement while maintaining competitive performance.

Keywords: Portfolio Optimization, Carbon Constraints, Climate Finance, Tracking Error, Minimum-Variance Portfolio, Net-Zero Trajectories, Three-Dimensional Trade-Off, Sustainable Investing, Paris Alignment

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1 Introduction

As global temperatures continue their inexorable rise the sixth–assessment synthesis of the Intergovernmental Panel on Climate Change (IPCC)¹ leaves little doubt: without a rapid and permanent re-direction of capital flows, the remaining carbon budget consistent with a $\leq 1.5^\circ\text{C}$ rise in global mean temperature will be exhausted well before 2030, making the more lenient 2°C guard-rail increasingly tenuous.¹ For institutional investors, this scientific deadline translates into a strategic imperative: portfolios are increasingly exposed to transition, liability, and physical risks that may resist diversification.

Yet fiduciary law obliges the same investors to maximise risk-adjusted returns for their beneficiaries. The perceived incompatibility between these two objectives has spawned what practitioners call the “sustainability tax”—the idea that any serious decarbonisation mandate must depress performance. Recent general-equilibrium models cast doubt on that heuristic. Pastor, Stambaugh, and Taylor² show that if climate risk is at least partially priced, lower-carbon firms should earn (on average) a *lower* cost of capital without jeopardising the Sharpe ratio of a well-diversified portfolio. Empirical work by Bolton and Kacperczyk³ reaches a complementary conclusion: equities that exhibit high historical emissions already command a “carbon premium” that resembles a priced factor rather than a free lunch. Taken together, these papers suggest that climate alignment is not necessarily purchased at the expense of performance—but neither study tackles the concrete optimisation choices that chief investment officers must make when carbon objectives collide with benchmark or policy constraints.

Our contribution unfolds along three lines. First, we extend the classical mean–variance framework conceptually to a $(\mu, \sigma, \text{carbon})$ setting, integrating financial performance and carbon intensity explicitly into the investment decision process. Unlike ESG scores, which aggregate multiple non-financial criteria, our analysis isolates financed emissions as a standalone optimisation axis. Although we do not numerically construct the full three-dimensional efficient frontier, we empirically illustrate how key portfolio archetypes relate expected returns, risk, and carbon footprint reductions using a representative universe of over 600 North-American equities spanning 2014–2024. Second, we re-interpret *tracking error*—the central risk metric in delegated asset management—as a fungible “budget” available for climate ambition. Building on the intuition of Andersson, Bolton, and Samama⁴, we quantify the opportunity cost of each incremental tonne of avoided CO₂e in terms of benchmark deviation. Finally, we empirically examine the performance of a *science-based* portfolio rule designed to reduce financed emissions by 10 % per annum, reflecting the Net-Zero pathways advocated by Dietz et al.⁵. To ensure the reliability of these insights, we also conduct sensitivity analyses across different estimation horizons, confirming the robustness of our conclusions to varying look-back windows and potential estimation noise.

Methodological transparency is central to our contribution. In addressing the multifaceted challenges of climate-conscious portfolio construction, we prioritise clarity in the definition of objectives, constraints, and trade-offs. Rather than framing decarbonisation as a binary choice between sustainability and performance, we adopt a quantitative perspective that explicitly models the tension between financial risk, expected return, and carbon exposure. This approach ensures that the implications of each allocation decision are both measurable and interpretable within a coherent risk-return framework.

More importantly, we view transparency not just as procedural diligence, but as a foundation for informed capital allocation amid structural change. As investors increasingly grapple with the dual imperatives of fiduciary responsibility and climate alignment, there is a pressing need for frameworks that render these objectives commensurable—allowing stakeholders to assess, with analytical precision, the opportunity costs associated with varying degrees of climate ambition.

Accordingly, we emphasise performance metrics most relevant to practitioner concerns, such as tracking error relative to established benchmarks and the magnitude of financed emissions reductions. By doing so, we seek to bridge the gap between high-level climate commitments and their concrete operationalisation within portfolio management processes. This dual lens—financial and environmental—forms the backbone of our empirical investigation, to which we now turn. To make these trade-offs tangible, we systematically analyse a set of prototypical portfolio strategies, each embodying a distinct resolution of the tension between decarbonisation ambition and benchmark fidelity.

Empirically, we evaluate five archetypes that span the practical design space confronted by asset owners:

- 1) the **unconstrained** value-weighted benchmark $P^{(vw)}$;
- 2) the **unconstrained** minimum-variance portfolio $P_{oos}^{(mv)}$;
- 3) its **static 50 % carbon-capped** sibling $P_{oos}^{(mv)}(0.5)$;
- 4) a **benchmark-neutral, tracking-error-minimising** design that also enforces a 50 % footprint cut, $P_{oos}^{(vw)}(0.5)$;
- 5) and a **dynamic Net-Zero** allocation $P_{oos}^{(vw)}(\text{NZ})$ that must follow a 10 % p.a. decline trajectory.

Together, these strategies allow us to ask two policy-relevant questions. *How expensive, in tracking-error units, is a given level of decarbonisation? And can a science-based emissions path be met without surrendering the equity risk premium?*

The remainder of the paper proceeds as follows. Section 2 describes the construction, filtering, and harmonisation of the North-American dataset, emphasising missing-value treatment and bankruptcy coding. Section 3 sets out the optimisation problems, carbon metrics, and performance statistics. Results are presented in Section 4, followed by an in-depth discussion of their theoretical and practical import in Section 5. Section 6 concludes with a reflection on transparency, policy relevance, and avenues for future research.

2 Data

In this section, we present the steps we took to filter the data from the North-American region, with the aim of harmonising the datasets and obtaining a consistent pool of companies per year that is comparable across every stage of our portfolio analysis.

2.1 Datasets

- `static`: descriptive information for 2,565 companies (ISIN code, company name, country and region).
- `scope1` and `scope2`: annual CO₂ emissions for direct and indirect emissions, respectively (in tonnes of CO₂).
- `yrevenue`: companies' annual revenue (in thousands of USD, normalized in millions of USD).
- `ycap`: companies' annual market capitalization (in millions of USD).
- `mcap`: companies' monthly market capitalization (in millions of USD).
- `yprice`: total-return index (price + dividends) at year-end (in USD).
- `mprice`: total-return index (price + dividends) at month-end (in USD).

The capitalization and index series run from the end of 1999 to the end of 2024, whereas the carbon data (emissions and revenue) span from the end of 2013 to the end of 2023. Restricting the study to firms domiciled in North America reduces the universe from 2,565 entities to the 671 companies.

2.2 Dataset Processing

Year	Original	Initial	Intermediate	Final	Removed in Backtest	Backtest Filter Ratio (%)	Bankruptcies	Bankruptcy Rate (%)
2014	671	656	532	526	6	98.87	0	0
2015	671	656	572	563	9	98.43	0	0
2016	671	656	584	578	6	98.97	0	0
2017	671	656	594	587	7	98.82	0	0
2018	671	656	601	596	5	99.17	0	0
2019	671	656	615	609	6	99.02	0	0
2020	671	656	625	618	7	98.88	0	0
2021	671	656	633	626	7	98.89	0	0
2022	671	656	647	641	6	99.07	11	1.72
2023	671	656	654	635	19	97.09	7	1.10
2024	671	656	655	630	25	96.18	10	1.59

Table 1: Annual sample sizes at each data-processing stage; firms removed by the backtest filter (Backtest Filter Ratio % = Final / Intermediate × 100); and bankruptcy counts and rates (Bankruptcy Rate % = Bankruptcy / Final × 100) for 2014–2024

2.2.1 Initial Processing

In the first phase of our analysis, we performed a rigorous preprocessing of the annual variables `yrevenue`, `scope1`, `scope2` and `ycap` to optimise the integrity and coherence of the data. This procedure was structured around three sequential methodological steps. Missing values at the beginning of the series were left unchanged, as they generally correspond to the period before the company's initial public offering. For any other missing observation—whether occurring between two available annual observations or at the end of the sample—we carried forward the last non-missing observation in the series (i.e., the most recent available value).

After this imputation step, we carried out a systematic analysis to identify columns containing only missing values across the entire set of data matrices. The objective was to isolate companies for which no exploitable information was available, in order to exclude them from subsequent analysis.

Finally, we removed from all our datasets the columns identified as entirely missing, thereby reducing our initial sample from 671 to 656 North-American companies. This 2.2% reduction in the sample represents an acceptable compromise between maximising the number of firms analysed and maintaining data quality (see columns Original and Initial in Table 1).

It should be noted that our imputation approach—relying solely on the last available observation—could theoretically bias the timing of corporate bankruptcy detection. This limitation is mitigated in subsequent Section 2.2.3 by intersecting the annual dataset with higher-frequency monthly data, which enables more precise identification of the abrupt discontinuities characteristic of corporate failures.

2.2.2 Intermediate Processing

As part of the intermediate data processing, we applied a stringent filtering procedure designed to retain only those companies for which complete data were available for the four essential annual variables: Scope 1 emissions (`scope1`), Scope 2 emissions

(scope2), annual revenue (y_{revenue}) and market capitalization (y_{cap}). Companies missing any of these variables in a given year were excluded to prevent gaps in critical financial and environmental measures.

Our filtering process rests on these four fundamental criteria applied systematically to each observation year. First, Scope 1 emission data must be non-missing, representing direct emissions from operations. Second, Scope 2 emission data must likewise be available, covering indirect emissions related to energy consumption. Third, annual revenue data must be present to enable accurate carbon-intensity calculations. Finally, market capitalization figures must be non-missing to avoid computational issues in portfolio weighting and risk assessments.

The application of these demanding criteria enabled us to build, for each year, a subset of companies with complete and reliable annual data. Table 1 illustrates the evolution of valid companies over the study period (see columns Initial and Intermediate). Although this filtering substantially reduces the annual sample size, it ensures the consistency and quality of the data used in subsequent steps.

Detailed monitoring of excluded companies and specific reasons for exclusion (missing emissions, unavailable capitalization, etc.) confirms that no systematic bias was introduced into our analytical sample. Finally, we intersected this filtered annual dataset with the monthly data, carrying forward only the companies validated at the annual level and retaining all their monthly observations for high-frequency analyses. In this step, companies were retained for the 120-month estimation window and in the 12-month test window.

2.2.3 Final Processing

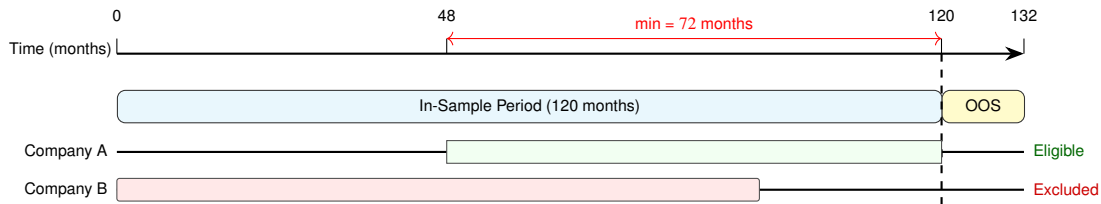


Figure 1: Diagram illustrating the company-filtering methodology based on the availability of monthly data. The in-sample estimation window spans 120 months, followed by a 12-month out-of-sample (OOS) test period. Within the estimation window, the $L = 72$ -month minimum continuous-history criterion ensures robust parameter estimates while mitigating survivorship bias.

After applying the filtering procedures to the annual variables described above, we now turn to the construction of the back-testing windows that will be used to evaluate the different investment strategies. This step draws primarily on the monthly returns derived from `mprice`, for which we performed rigorous preprocessing.

Preparation of monthly returns First, we identified and only treated isolated missing values in the monthly price series (`mprice`) by carrying forward the last non-missing observation—when the gap occurred between two valid prices—to maintain consistency with our annual variable treatment. We then computed the monthly returns and deliberately left any residual missing values unfilled, thereby preventing the propagation of potential estimation errors. Finally, we aligned our investment universe with that obtained after the secondary (annual) filtering, retaining only the returns of those companies that had been deemed eligible in the annual filter. This consistent treatment across annual and monthly data ensures the homogeneity of our analysis sample.

Temporal structure of the back-testing windows For each test period we define a strict temporal structure comprising

- an *in-sample* estimation period of 10 years (120 months), and
- an *out-of-sample* test period of 1 year (12 months).

Our methodology imposes a fundamental constraint: at least $L = 72$ months of *continuous* return history must be available at the *end* of the estimation period (see Figure 1). This minimum-history requirement applies specifically to the last part of the estimation window, which allows us to include firms with incomplete data at the beginning of the period. Thus, younger companies or firms that began reporting data mid-period can be incorporated into the analysis provided they satisfy the 72-consecutive-month criterion.

While the final-segment requirement of 72 continuous months supplies enough data for reliable estimation, we deliberately impose no minimum on the first 48 months of the 120-month window. This asymmetry reflects a trade-off between depth of history and breadth of coverage: insisting on a full 120-month record would systematically exclude younger listings, whereas relaxing the constraint on the initial segment allows recently floated firms to enter the sample without compromising the six-year data span needed for stable parameter estimates.

The adopted filtering methodology offers several significant advantages. By requiring a data history only for the last 72 months of the estimation period (rather than for the entire 120 months), it greatly mitigates survivorship bias, which would otherwise favour older firms. The approach also provides balanced treatment of delistings, avoiding both the excessive exclusion of firms that experience temporary interruptions and the inclusion of those whose data are too fragmented for reliable estimation.

Following this first phase of filtering, we carried out a harmonisation process to guarantee coherence between the different temporal dimensions of the analysis. Superimposing the monthly criteria on the initial annual filter yields a homogeneous set of firms for each study year. This harmonisation mechanism ensures that the same companies are represented consistently across all datasets, eliminating disparities that could bias portfolio construction. As shown in Table 1 (see column Final)¹, the process results in a stable number of retained firms for every study year, thereby creating a solid foundation for the subsequent analyses. The resulting stability of the corporate panel is essential for ensuring the comparability of results across the different periods and portfolio strategies examined. After processing the isolated missing values and merging at the Final step with the market-capitalization series `mcap`, we confirmed that `mcap` contained no missing data for all periods in which it is applied.

Complementing the processing of the annual financial data (`ycap`, `yrevenue`), this monthly filter naturally excludes firms whose values may have been incorrectly imputed in earlier stages, thereby further enhancing the overall quality of the sample. Moreover, the criterion is fully consistent with the filters applied to the annual data, ensuring methodological coherence throughout. Finally, the 72-month requirement strikes an optimal balance between sample diversification and the statistical robustness of the parametric estimates needed to construct the portfolios.

All subsequent estimations therefore rely on a minimum of $L = 72$ consecutive monthly returns—a compromise that preserves a broad universe while containing the sampling noise of the covariance matrix. Section 4.6 shows that our results are virtually unchanged when L is widened to 84 m or shortened to 60 m and 48 m.¹

2.2.4 Bankruptcies Processing

Our study employs a systematic methodology for handling corporate defaults in the portfolio back-tests, thereby enhancing the robustness of the evaluations while preserving the realism of the simulations. This standardised approach avoids biases linked to asset disappearances and maintains a faithful representation of performance—an aspect that is particularly crucial when comparing environmentally constrained investment strategies whose sectoral diversification may be affected. The methodology identifies five default configurations in the monthly return series, all treated according to a single coherent rule :

1. Zero-only tail: a stretch of zeros; the first zero is recoded as -100% , the remainder stay zero.
2. Missing-only tail: a sequence of NAs; the first NA becomes -100% , the rest are set to zero.
3. Explicit -100% present: when a -100% return already appears, that observation marks default; subsequent values are forced to zero.
4. Valid returns followed by NAs: the first NA is recoded as -100% , later NAs become zero.
5. Alternating -100% and NAs: the earliest problematic point (-100% or NA) defines default; everything afterwards is zero.

This structured and unified rule guarantees a single, time-stamped default per series and eliminates spurious post-default returns. It proved especially relevant for the years 2022–2024, during which we detected all 28 default events observed in our sample (See Table 1).

This method confers four major advantages. First, it preserves temporal information by precisely recording the default date. Second, the standardisation (-100% at default, then zeros) ensures continuity in portfolio evaluation without requiring *ad-hoc* adjustments during the test period. Third, the assumption of total loss at default reflects the conservative stance generally adopted in credit-risk analysis. Finally, converting the post-default missing data to zeros eliminates statistical artefacts that might otherwise bias performance and volatility measures. This coherent procedure successfully handled every default event over the 2014–2024 period, with an annual frequency consistent with historical statistics for the North-American market.

2.2.5 Rationale for Monthly Returns

After applying our initial filtering and bankruptcy exclusions, `yprice` presented no missing values. Nevertheless, we opted for monthly returns for two key reasons:

1. **Statistical power and precision.** By using 12 data points per year instead of one, we reduce the standard error of estimated moments (mean, variance, skewness) by roughly $\sqrt{12}$, enabling more reliable risk estimates, factor loadings, and backtests—especially critical in cross-sectional and time-series regressions.
2. **Industry practice and model alignment.** Most academic (e.g. Fama–French^{6,7}) and professional factor models, as well as risk-management systems, operate at monthly frequency. Monthly returns strike the optimal balance between noise (too high at daily) and information loss (too high at annual), ensuring comparability and integration with widely used benchmarks and analytics.

To preserve intra-year granularity—essential for accurately capturing dividend flows and market shocks—we therefore deliberately prioritised monthly returns calculated from `mprice`, while maintaining `yprice` on hand for any further annual-level analyses.

¹A shorter window adds estimation noise but limits survivorship bias, whereas a longer one does the opposite.

3 Methodology

All strategies are implemented over rolling windows spanning 2014 through 2024, as per the indexing convention in Section 3.6.

3.1 Covariance Estimation with Incomplete Histories

For each calendar year Y , we define a rolling ten-year estimation window $\mathcal{T}_{Y+1} = [Y - 9, Y]$, which comprises the 120 monthly returns from January of year $Y - 9$ through December of year Y (see Section 2). By design, any pair of stocks retained in the universe has at least 72 overlapping observations within $\mathcal{T}_{Y+1}$. First, we define the historical mean estimator for each asset: $\hat{\mu}_{i,Y+1} = \frac{1}{|T_{ij}|} \sum_{t \in T_{ij}} R_{i,t}$.

Pairwise Sample Covariances. For each pair of assets (i, j) , let $T_{ij} \subseteq \mathcal{T}_{Y+1}$ denote the months in which *both* returns are observed (so $|T_{ij}| \geq 72$). We then compute the pairwise covariance estimator:

$$\tilde{\sigma}_{ij,Y+1} = \frac{1}{|T_{ij}|} \sum_{t \in T_{ij}} (R_{i,t} - \hat{\mu}_{i,Y+1})(R_{j,t} - \hat{\mu}_{j,Y+1}),$$

where $\hat{\mu}_{i,Y+1}$ and $\hat{\mu}_{j,Y+1}$ are the sample means over T_{ij} .

We employ this pure pairwise estimator exclusively for all portfolios and do not apply any subsequent numerical correction (e.g. resetting of small negative eigenvalues). Empirically, the resulting covariance matrices remain sufficiently positive semi-definite for all portfolio optimizations in this study. All routines are implemented in Python, using `pandas`⁸ for data handling and SciPy's SLSQP solver⁹ for optimization.

3.2 Unconstrained Portfolios

3.2.1 Minimum-Variance Portfolio

We construct the out-of-sample minimum-variance portfolio $P_{\text{mos}}^{(mv)}$ by solving, at each annual rebalance date (last trading day of December Y , $Y = 2013, \dots, 2023$),

$$\min_{\alpha_Y} \alpha_Y^\top \Sigma_{Y+1} \alpha_Y \quad \text{s.t.} \quad \alpha_Y^\top \mathbf{1} = 1, \quad \alpha_Y \geq 0.$$

Denoting the vector of asset returns in month t by $R_t = (R_{1,t}, \dots, R_{N,t})^\top$, we apply the weights α_Y over the subsequent twelve months. In each month $t = Y + 1, \dots, Y + 12$, the portfolio return is $R_{p,t} = \alpha_{t-1}^\top R_t$, and the weights evolve by full reinvestment: $\alpha_{i,t} = \frac{\alpha_{i,t-1}(1+R_{i,t})}{1+R_{p,t}}$, $i = 1, \dots, N$.

Notably, for the first month after the rebalance ($t = Y + 1$), the portfolio return reduces to: $R_{p,Y+1} = \sum_{i=1}^N \alpha_{i,Y} R_{i,Y+1}$, i.e. the dot product of the weights set at 31 Dec Y with the asset returns of Jan $Y + 1$.

This update rule ensures that $\sum_i \alpha_{i,t} = 1$ at all times and that each weight automatically increases when the corresponding asset outperforms the portfolio and decreases otherwise.

3.2.2 Value-Weighted Benchmark Portfolio

We construct the value-weighted benchmark portfolio $P^{(vw)}$ by updating the asset weights at each monthly rebalance date—namely, the last trading day of month t —as $w_{i,t} = \frac{\text{Cap}_{i,t}}{\sum_{j=1}^N \text{Cap}_{j,t}}$, where $\text{Cap}_{i,t}$ is the market capitalization of asset i at month-end t (in USD), ensuring $\sum_{i=1}^N w_{i,t} = 1$. The portfolio return in the following month is then given by $R_{t+1}^{(vw)} = \sum_{i=1}^N w_{i,t} R_{i,t+1}$. In practice, each weight $w_{i,t}$ is computed using the market-cap on the last trading day of month t , so that—for example—the weight used for the January $Y + 1$ return is formed from the December Y capitalizations.

3.3 Carbon Intensity and Portfolio Footprint

For each firm i and year Y , we define carbon intensity as

$$CI_{i,Y} = \frac{E_{i,Y}}{Revenue_{i,Y}} \quad (\text{tCO}_2\text{e/MUSD}),$$

where $E_{i,Y}$ are total annual emissions (scope 1+ scope 2) in tonnes of CO₂-equivalent and $Revenue_{i,Y}$ is annual revenue in million USD (see Section 2).

Weighted Average Carbon Intensity (WACI). For a portfolio p with weights $\alpha_{i,Y}$ at year Y ,

$$WACI_Y^{(p)} = \sum_{i=1}^N \alpha_{i,Y} CI_{i,Y}.$$

Carbon Footprint (CF). Let $V_Y = \sum_{i=1}^N V_{i,Y}$ be portfolio value and $V_{i,Y} = \alpha_{i,Y} V_Y$ the dollar amount in firm i . The ownership share is $o_{i,Y} = V_{i,Y}/Cap_{i,Y}$, with $Cap_{i,Y}$ the firm's market cap. Then

$$CF_Y^{(p)} = \frac{1}{V_Y} \sum_{i=1}^N o_{i,Y} E_{i,Y}.$$

We compute $CF_Y^{(P_{\text{OOS}}^{(+)})}$ for each year, assuming a starting wealth $V_{2013} = 1\text{MUSD}$.

3.4 Constrained Strategies

3.4.1 Minimum-Variance Portfolio under 50% Carbon Constraint

We denote by $P_{\text{OOS}}^{(mv)}(0.5)$ the out-of-sample minimum-variance portfolio subject to a 50% carbon footprint constraint. At each annual rebalance date (last trading day of December Y , $Y = 2013, \dots, 2023$), the weights $\alpha_Y^{(0.5)}$ solve

$$\min_{\alpha_Y} \alpha_Y^\top \Sigma_{Y+1} \alpha_Y \quad \text{s.t.} \quad \alpha_Y^\top \mathbf{1} = 1, \quad \alpha_Y \geq 0, \quad CF_Y^{(p)} \leq 0.5 CF_Y^{(P_{\text{OOS}}^{(mv)})},$$

where $CF_Y^{(p)} = \sum_{i=1}^N \alpha_{i,Y} CF_{i,Y}$ is the carbon footprint of the candidate portfolio and $CF_Y^{(P_{\text{OOS}}^{(mv)})}$ that of the unconstrained minimum-variance benchmark. We then apply the same monthly reinvestment update rule as in Section 3.2.1.

3.4.2 Tracking-Error Minimization under 50% Carbon Constraint

We denote by $P_{\text{OOS}}^{(TE)}(0.5)$ the out-of-sample tracking-error minimizing portfolio subject to a 50% carbon footprint constraint. At each annual rebalance date (last trading day of December Y , $Y = 2013, \dots, 2023$), the weights $\alpha_Y^{(TE)}$ solve

$$\min_{\alpha_Y} (\alpha_Y - \alpha_Y^{(vw)})^\top \Sigma_{Y+1} (\alpha_Y - \alpha_Y^{(vw)}) \quad \text{s.t.} \quad \alpha_Y^\top \mathbf{1} = 1, \quad \alpha_Y \geq 0, \quad CF_Y^{(p)} \leq 0.5 CF_Y^{(P^{(vw)})},$$

where $\alpha_Y^{(vw)}$ are the value-weighted weights defined in Section 3.2.2, $CF_Y^{(p)} = \sum_{i=1}^N \alpha_{i,Y} CF_{i,Y}$ is the portfolio's carbon footprint, and $CF_Y^{(P^{(vw)})}$ that of the value-weighted benchmark. We then apply the same monthly reinvestment update rule as in Section 3.2.1.

3.4.3 Dynamic 10% Annual Decarbonization Strategy

We denote by $P_{\text{OOS}}^{(vw)}(NZ)$ the out-of-sample value-weighted portfolio following a dynamic 10% annual decarbonization trajectory. At each annual rebalance date (last trading day of December Y , $Y = 2013, \dots, 2023$), the weights $\alpha_Y^{(NZ)}$ solve

$$\min_{\alpha_Y} (\alpha_Y - \alpha_Y^{(vw)})^\top \Sigma_{Y+1} (\alpha_Y - \alpha_Y^{(vw)}) \quad \text{s.t.} \quad \alpha_Y^\top \mathbf{1} = 1, \quad \alpha_Y \geq 0, \quad CF_Y^{(p)} \leq (1 - \theta)^{Y-Y_0+1} CF_{Y_0}^{(P^{(vw)})},$$

where $\theta = 0.10$ is the annual decarbonization rate, $Y_0 = 2013$ is the base year, $CF_Y^{(p)} = \sum_{i=1}^N \alpha_{i,Y} CF_{i,Y}$ is the portfolio's carbon footprint at year Y , and $CF_{Y_0}^{(P^{(vw)})}$ that of the value-weighted benchmark in the base year. We then apply the same monthly reinvestment update rule as in Section 3.2.1.

3.5 Performance Metrics

3.5.1 Definition of Out-of-Sample Returns and Cumulative Returns

We evaluate the performance of our portfolio strategies using out-of-sample returns. For each portfolio p , the monthly out-of-sample return at time t is computed as: $R_{p,t} = \sum_{i=1}^N \alpha_{i,t-1} \cdot R_{i,t}$ where $\alpha_{i,t-1}$ represents the portfolio weight of asset i at the end of month $t-1$, and $R_{i,t}$ is the return of asset i during month t .

The cumulative return of portfolio p up to time T is given by: $C_{p,T} = \prod_{t=1}^T (1 + R_{p,t}) - 1$

3.5.2 Definition of Comparative Statistics

We employ several metrics to compare portfolio strategies:

- **Annualized Return:** $\mu_p^{(ann)} = (1 + \mu_p)^{12} - 1$, where μ_p is the monthly average return.
- **Annualized Volatility:** $\sigma_p^{(ann)} = \sigma_p \cdot \sqrt{12}$, where σ_p is the monthly standard deviation of returns.
- **Sharpe Ratio:** $SR_p = \frac{\mu_p^{(ann)} - r_f}{\sigma_p^{(ann)}}$, where r_f is the annualized risk-free rate.
- **Maximum Drawdown:** $MDD_p = \min_t \left(\frac{C_{p,t}}{\max_{s \leq t} C_{p,s}} - 1 \right)$, measuring the largest peak-to-trough decline.

Tracking error: The annualized standard deviation of the return difference $R_p - R_b$, formally $TE = \sqrt{\text{Var}(R_p - R_b)}$.

Additionally, we compute the excess return as the difference between the portfolio return and the risk-free rate, allowing for risk-adjusted performance assessment across different market environments.

3.6 Timing conventions for weights, returns and carbon data

To keep methodology, results and project briefs consistent, we fix the following calendar:

- 1) The evaluation window runs from 2014 to 2024. We index full calendar years by $Y \in \{2014, \dots, 2024\}$.
- 2) Just after the market close of 31 Dec ($Y-1$) we decide the weight vector $\alpha_{i,Y}$; it is held unchanged from Jan Y to Dec Y .
- 3) Carbon data—scope-1 + scope-2 emissions $E_{i,\tau}$, carbon intensity $CI_{i,\tau}$, revenues *etc.*—refer to fiscal year τ and become public in spring $\tau+1$. **Hence the optimiser only knows $CI_{i,Y-1}$ when it sets $\alpha_{i,Y}$.**
- 4) Monthly index $t = 1, \dots, 132$ counts from Jan 2014; we denote by $Y(t)$ the calendar year of month t .

Dictionary of symbols

Symbol / object	Index	Date reference	Interpretation
$\alpha_{i,Y}$	Y	31 Dec ($Y-1$)	Weight held during year Y
$R_{p,t}$	t	Month t	OOS return using $\alpha_{i,Y(t)}$
$CI_{i,\tau}, E_{i,\tau}, \text{Revenue}_{i,\tau}$	τ	Fiscal year τ	Raw climate / financial inputs
$WACI_Y^{(p)}, CF_Y^{(p)}$	Y	Calendar year Y	Metrics with $\alpha_{i,Y}$ & $CI_{i,Y}$

Table 2: Time grid adopted for 2014–2024.

Core equations

$$\begin{aligned}
 R_{p,t} &= \sum_{i=1}^N \alpha_{i,Y(t)} R_{i,t}, \quad t = 1, \dots, 132; \\
 CF_Y^{(p)} &= \sum_{i=1}^N \alpha_{i,Y} CI_{i,Y}; \\
 WACI_Y^{(p)} &= \frac{\sum_{i=1}^N \alpha_{i,Y} CI_{i,Y}}{\sum_{i=1}^N \alpha_{i,Y} V_{i,Y}}, \quad V_{i,Y} = \text{market-cap at 31 Dec } (Y-1).
 \end{aligned} \tag{1}$$

Constraint evaluation at 31 Dec ($Y-1$) The optimiser *only* has access to $CI_{i,Y-1}$, therefore

$$\begin{aligned}
 \sum_i \alpha_{i,Y} CI_{i,Y-1} &\leq 0.5 \sum_i \alpha_{i,Y}^{(mv)} CI_{i,Y-1}, \\
 \sum_i \alpha_{i,Y} CI_{i,Y-1} &\leq (1 - 0.10)^{Y-2013} CF_{2013}^{(P^{(vw)})}.
 \end{aligned}$$

After fiscal year Y closes, we compute realised $CF_Y^{(p)}$ and $WACI_Y^{(p)}$ with the fresh $CI_{i,Y}$, but *without retro-trading*. Thus performance graphs (Section 4) remain correctly dated and need no re-indexing.

4 Results

Unless stated otherwise, all risk–return and carbon metrics are computed with the baseline $L = 72$ m (See Section 2.2.3); Section 4.6 checks the robustness of every finding to $L = 48, 60$, and 84 m.

4.1 Unconstrained Portfolios

4.1.1 Minimum-Variance Portfolio

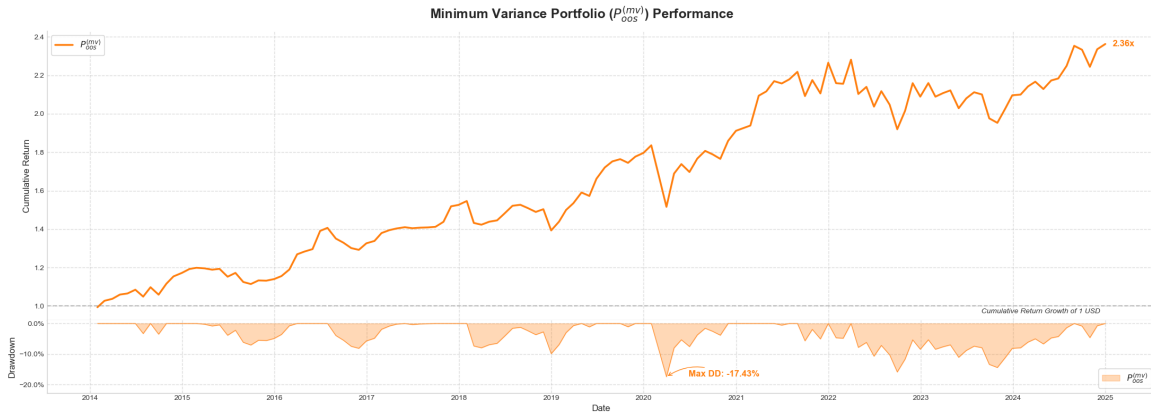


Figure 2: Cumulative performance and drawdown of the out-of-sample minimum-variance portfolio $P_{OOS}^{(mv)}$ in the North American equity market, 2014–2024^(3,6). **Top panel:** growth of 1 USD invested at the start of 2014. **Bottom panel:** corresponding drawdown series, with a maximum peak-to-trough decline of 17.43% in March 2020.

Metric	$P_{OOS}^{(mv)}$
Annualized Average Return	0.08553
Annualized Volatility	0.11949
Annualized Sharpe Ratio	0.58634
Annualized Excess Return	0.07026
Annualized Cumulative Return	0.08130
Monthly Average Return	0.00713
Monthly Volatility	0.03450
Monthly Sharpe Ratio	0.16926
Monthly Excess Return	0.00585
Maximum Monthly Return	0.11404
Minimum Monthly Return	-0.10116
Maximum Drawdown	-0.17426
Maximum Drawdown Date	2020-03

Table 3: Out-of-sample performance metrics for the ($P_{OOS}^{(mv)}$) in the North America equity universe over the 2014–2024 period^(3,6).

As summarized in Table 3, the $P_{OOS}^{(mv)}$ delivered an annualized average return of 8.55% over the 2014–2024 period in the North American equity market, with an associated annualized volatility of 11.95%. This trade-off yields a Sharpe ratio of 0.59, indicating a favourable risk-adjusted profile relative to typical market benchmarks. The maximum drawdown of 17.4% occurred in March 2020 amidst the global COVID-19 turmoil, yet the portfolio fully recouped losses within nine months, thereby demonstrating notable resilience in the face of severe market stress (see Figure 2).

On a monthly horizon, the $P_{OOS}^{(mv)}$ realized an average return of 0.71% against a volatility of 3.45%, reinforcing the strategy’s capacity to generate steady returns with controlled downside. The relatively low dispersion between maximum and minimum monthly returns (11.40% versus −10.12%) further attests to effective diversification across sectors and industries. Taken together, these results substantiate the $P_{OOS}^{(mv)}$ ’s role as a robust defensive allocation in North American equities, capable of cushioning against extreme shocks while delivering consistent performance.

4.1.2 Value-Weighted Benchmark Portfolio

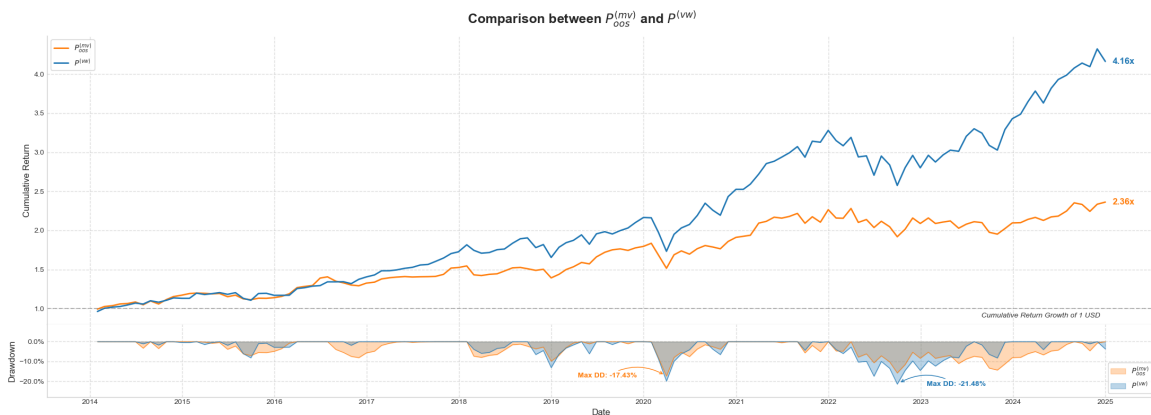


Figure 3: Comparison of cumulative performance and drawdown for the $P_{OOS}^{(mv)}$ (orange) and the value-weighted benchmark $P^{(vw)}$ (blue) in the North American equity universe, 2014–2024^(3,6). **Top panel:** growth of 1 USD invested at the start of 2014. **Bottom panel:** drawdown series, highlighting maximum peak-to-trough declines of 21.48% for $P^{(vw)}$ (September 2022) and 17.43% for $P_{OOS}^{(mv)}$ (March 2020).

Metric	$P_{oos}^{(mv)}$	$P^{(vw)}$
Annualized Average Return	0.08553	0.14073
Annualized Volatility	0.11949	0.14452
Annualized Sharpe Ratio	0.58634	0.86767
Annualized Excess Return	0.07026	0.12546
Annualized Cumulative Return	0.08130	0.13842
Monthly Average Return	0.00713	0.01173
Monthly Volatility	0.03450	0.04172
Monthly Sharpe Ratio	0.16926	0.25047
Monthly Excess Return	0.00585	0.01046
Maximum Monthly Return	0.11404	0.12550
Minimum Monthly Return	-0.10116	-0.12347
Maximum Drawdown	-0.17426	-0.21480
Maximum Drawdown Date	2020-03	2022-09
Recovery Period (months)	9	11

Table 4: Key out-of-sample performance metrics for the $P_{oos}^{(mv)}$ and VW portfolios, 2014–2024.

upside capture during bull phases, exhibits less resilience during market troughs. This comparison illustrates the risk–return trade-off: $P^{(vw)}$ maximizes growth at the expense of heightened tail risk, while the $P_{oos}^{(mv)}$ delivers steadier performance with more controlled downside. A significant share of the benchmark’s out-performance after 2020 can be traced to the outsized run-up of mega-cap technology names, whose cumulative weights peaked above 25% of the index between 2020 and 2024.

4.2 Unconstrained Portfolios Carbon Intensity and Footprint

4.2.1 Unconstrained Portfolios Footprint

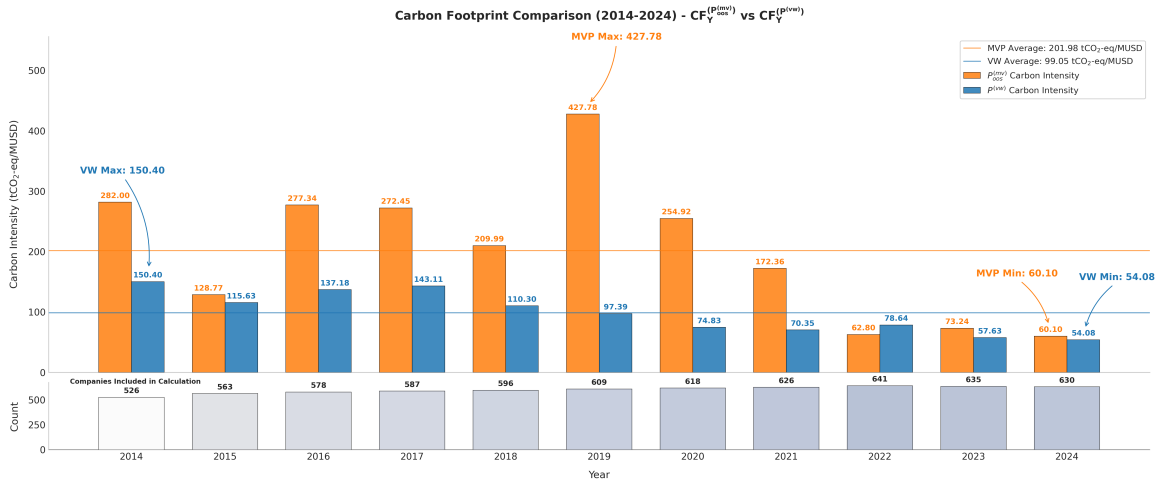


Figure 4: **Year-by-year weighted carbon footprint.** Bars report the annual carbon intensity (Scope-1 + Scope-2 emissions per million USD of market capitalisation, in tCO₂-eq/MUSD) of the minimum-variance portfolio $P_{oos}^{(mv)}$ (MVP) (orange) and of the value-weighted benchmark $P^{(vw)}$ (VW) (blue) between 2014 and 2024^(3,6). Horizontal lines give the sample-average footprints (202 tCO₂-eq/MUSD for $P_{oos}^{(mv)}$, 99 tCO₂-eq/MUSD for $P^{(vw)}$). Dotted grey columns at the bottom indicate the number of companies included in each year’s calculation. Arrows mark the yearly maxima and minima for both series.

Year	$CF_Y^{(P_{oos}^{(mv)})}$	$CF_Y^{(P^{(vw)})}$	$\Delta(\%)$	N_{firms}
2014	282.00	150.40	87.50	526
2015	128.77	115.63	11.36	563
2016	277.34	137.18	102.18	578
2017	272.45	143.11	90.38	587
2018	209.99	110.30	90.38	596
2019	427.78	97.39	339.25	609
2020	254.92	74.83	240.65	618
2021	172.36	70.35	144.99	626
2022	62.80	78.64	-20.15	641
2023	73.24	57.63	27.09	635
2024	60.10	54.08	11.12	630

Table 5: Annual carbon footprint $CF_Y^{(P^{(+)})}$ of the minimum-variance portfolio ($P_{oos}^{(mv)}$) and of the value-weighted benchmark ($P^{(vw)}$) between 2014 and 2024^(3,6), together with the percentage gap $\Delta = [CF^{(P_{oos}^{(mv)})} / CF^{(P^{(vw)})} - 1] \times 100$ and N the number of firms in the common investable universe.

Figure 3 juxtaposes the cumulative growth trajectories of the $P_{oos}^{(mv)}$ and the value weighted benchmark ($P^{(vw)}$), revealing that while the $P^{(vw)}$ portfolio outpaced the $P_{oos}^{(mv)}$, ending 2024 at approximately 4.16x its initial value versus 2.36x for the $P_{oos}^{(mv)}$, it also experienced more severe drawdowns (21.48% vs. 17.43%) and longer recovery periods (11 vs. 9 months). Table 4 quantifies these trade-offs: the VW strategy delivered higher annualized returns (14.08% vs. 8.55%) and a superior Sharpe ratio (0.87 vs. 0.59), but incurred greater volatility (14.45% vs. 11.95%) and deeper losses in stress episodes.

Despite its lower absolute return, the $P_{oos}^{(mv)}$ ’s more muted volatility and reduced drawdown profile underscore its appeal as a defensive allocation in the North American equity market. The faster recovery following the COVID-19 shock further highlights the $P_{oos}^{(mv)}$ ’s capacity to preserve capital under extreme conditions, whereas the VW benchmark, though superior in

The carbon footprint comparison in Fig. 4 and Table 5 reveals a persistent “carbon premium” embedded in the $P_{oos}^{(mv)}$. Except for 2022, $P_{oos}^{(mv)}$ exhibits a substantially higher emissions intensity than the value-weighted benchmark $P^{(vw)}$, averaging 202 tCO₂-eq/MUSD over 2014–2024—more than double the benchmark’s 99—and surging to 428 tCO₂-eq/MUSD in 2019, a gap of over 340 %. This systematic tilt arises because variance minimisation tends to overweight low-volatility sectors (e.g., utilities, energy, materials) that, in most North American markets, coincide with heavy emitters. Although both portfolios demonstrate a pronounced downward trajectory— $P_{oos}^{(mv)}$ from 282 to 60 and $P^{(vw)}$ from 150 to 54 between 2014 and 2024—the $P_{oos}^{(mv)}$ steeper decarbonisation (−79 % vs. −64 % for the benchmark) briefly reverses the gap in 2022, when an abrupt reduction in energy exposure drives the minimum-variance allocation below the benchmark.

Notably, the $P_{oos}^{(mv)}$'s carbon intensity also displays far greater year-to-year volatility than the benchmark's smoother decline. Small shifts in the covariance matrix can trigger sizeable reallocations toward or away from a handful of high-emission stocks, causing spikes in the $P_{oos}^{(mv)}$ footprint that are absent in the $P^{(vw)}$ series. This instability underscores the risk that an unconstrained, risk-efficient mandate can inadvertently amplify environmental exposure precisely when portfolio diversification would otherwise mitigate it. Introducing explicit carbon constraints—as in the tracking-error and decarbonisation strategies—can both cap overall emissions and dampen these volatility spikes, achieving a more stable and sustainable footprint with only a marginal reduction in risk-adjusted performance. Overall, the evidence underscores the importance of complementing risk-based allocation rules with climate criteria: without them, a minimum-variance mandate alone does *not* guarantee a low-carbon outcome.

4.2.2 Unconstrained Portfolios Weighted Average Carbon Intensity

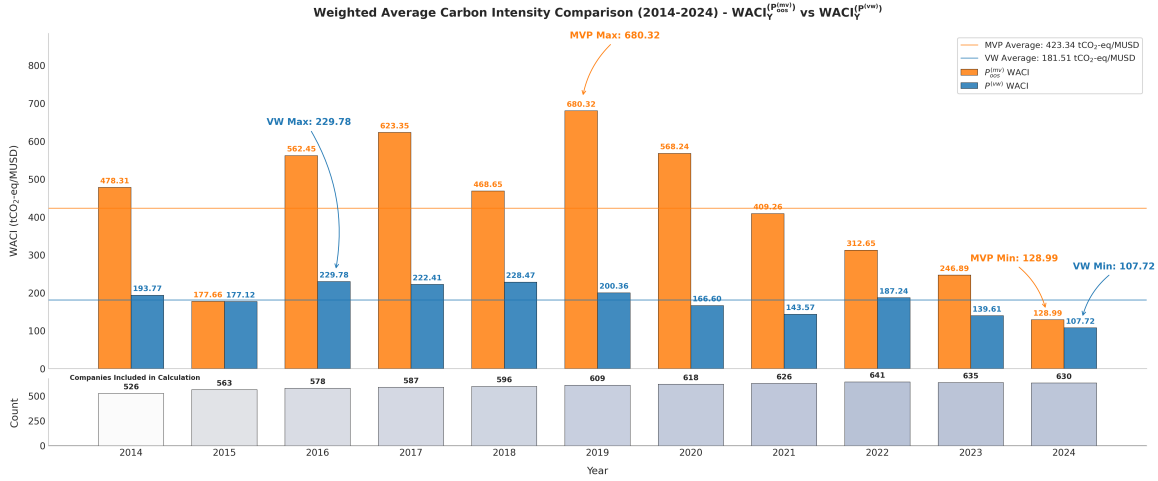


Figure 5: **Year-by-year weighted-average carbon intensity.** Each bar shows the portfolio-level WACI_Y^(P⁺)—Scope-1 + Scope-2 tonnes of CO₂-equivalent per million USD of market value—for the out-of-sample minimum-variance portfolio $P_{oos}^{(mv)}$ (MVP) (orange) and the value-weighted benchmark $P^{(vw)}$ (VW) (blue) from 2014 to 2024^(3,6). Horizontal lines mark sample averages (423 tCO₂/MUSD for $P_{oos}^{(mv)}$, 181 tCO₂/MUSD for $P^{(vw)}$). Grey columns (bottom axis) give the number of stocks used in each annual calculation. Arrows identify the sample maxima and minima for both series.

Year	WACI _Y ^(P^{mv})	WACI _Y ^(P^{vw})	Δ(%)	N _{firms}
2014	478.31	193.77	146.85	526
2015	177.66	177.12	0.30	563
2016	562.45	229.78	144.78	578
2017	623.35	222.41	180.27	587
2018	468.65	228.47	105.13	596
2019	680.32	200.36	239.54	609
2020	568.24	166.60	241.09	618
2021	409.26	143.57	185.06	626
2022	312.65	187.24	66.98	641
2023	246.89	139.61	76.84	635
2024	128.99	107.72	19.74	630

Table 6: Annual WACI_Y^(P⁺) of the minimum-variance portfolio ($P_{oos}^{(mv)}$) and of the value-weighted benchmark ($P^{(vw)}$) between 2014 and 2024^(3,6). Δ is the percentage excess of the $P_{oos}^{(mv)}$ over the benchmark, and N is the number of stocks in the common universe for that year.

The weighted-average carbon-intensity (WACI) profile of the unconstrained $P_{oos}^{(mv)}$ mirrors—but magnifies—the footprint evidence. Over 2014–2024 WACI^(P^{mv}) averages 423 tCO₂-eq/MUSD—more than double the benchmark's 182—and it tops the benchmark in every one of the eleven calendar years (Table 6). The gap is modest in 2015, when a rally in large, capital-intensive utilities briefly lifts WACI^(P^{vw}), but it balloons thereafter, touching +241 % in 2019–2020. Both portfolios decarbonise in absolute terms— $P_{oos}^{(mv)}$ declines from 478 to 129 tCO₂/MUSD, $P^{(vw)}$ from 194 to 108—yet $P_{oos}^{(mv)}$'s steeper fall compresses the *relative* premium rather than enlarging it. Even so, by 2024 $P_{oos}^{(mv)}$ still stands roughly 20 % above $P^{(vw)}$ and remains outside the upper bound that the Paris-aligned benchmark appendix¹⁰ of sets for a passive strategy in this region, underscoring the tension between pure variance minimisation and a 1.5 °C trajectory.

Two mechanisms drive this persistent disparity. First, portfolio-level variance is minimised by allocating to low-beta, high-dividend sectors—utilities, pipelines, telecoms—that are structurally emission-heavy in North America; in a value-weighted index these firms carry lower caps and therefore contribute less to WACI^(P^{vw}). Second, WACI normalises emissions by each firm's *own* revenue, amplifying the impact of companies with large absolute emissions but modest sales. Small shifts in the covariance matrix can therefore move $P_{oos}^{(mv)}$'s weight toward or away from a handful of such issuers, creating the pronounced year-to-year swings visible in Figure 5. The combination of a higher structural exposure and greater volatility makes the unconstrained $P_{oos}^{(mv)}$ environmentally fragile: without an explicit carbon constraint it fails to converge toward a Paris-compliant decarbonisation path, motivating the tracking-error and net-zero designs investigated in subsequent Sections 4.3.2 and 4.3.3.

and tends to come at the expense of mid-cap energy and utility names, replaced by lower-emission industrials and technology stocks. Because these substitutes exhibit similar low-beta profiles, overall risk is unaffected, explaining the near-identical drawdown pattern during the COVID-19 sell-off. The experiment therefore illustrates that a simple, hard constraint on absolute emissions can deliver a footprint without materially altering either the defensive risk characteristics or the growth trajectory of a minimum-variance allocation. In practice, investors seeking climate alignment need not relinquish the diversification benefits of risk-efficient portfolios; a carefully calibrated carbon budget suffices to reconcile the two objectives.

4.3.2 Value Weighted Benchmark Tracking-Error Minimisation under a 50% Carbon Budget

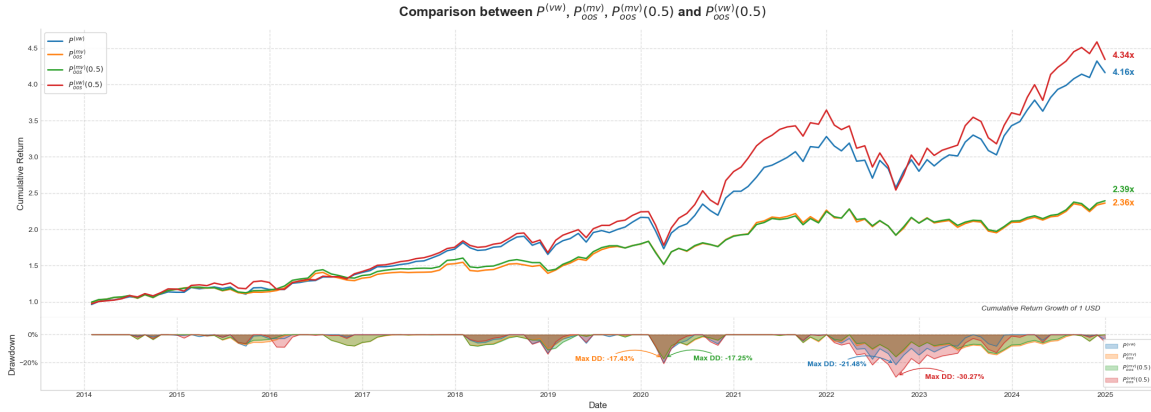


Figure 8: **Tracking-error portfolio versus unconstrained strategies.** Cumulative wealth (top) and drawdowns (bottom) for the value-weighted benchmark $P^{(vw)}$ (blue), the unconstrained minimum-variance portfolio $P_{oos}^{(mv)}$ (orange), the carbon-capped $P_{oos}^{(mv)}(0.5)$ (green), and the *tracking-error* portfolio $P_{oos}^{(vw)}(0.5)$ (red) between 2014-2024^(3,6), whose optimisation minimises return tracking-error to the benchmark while halving its footprint each year. By 2024 $P_{oos}^{(vw)}(0.5)$ compounds to 4.34x, outpacing the benchmark (4.16x) at the cost of a deeper maximum drawdown (-30.3 % vs. -21.5 %).

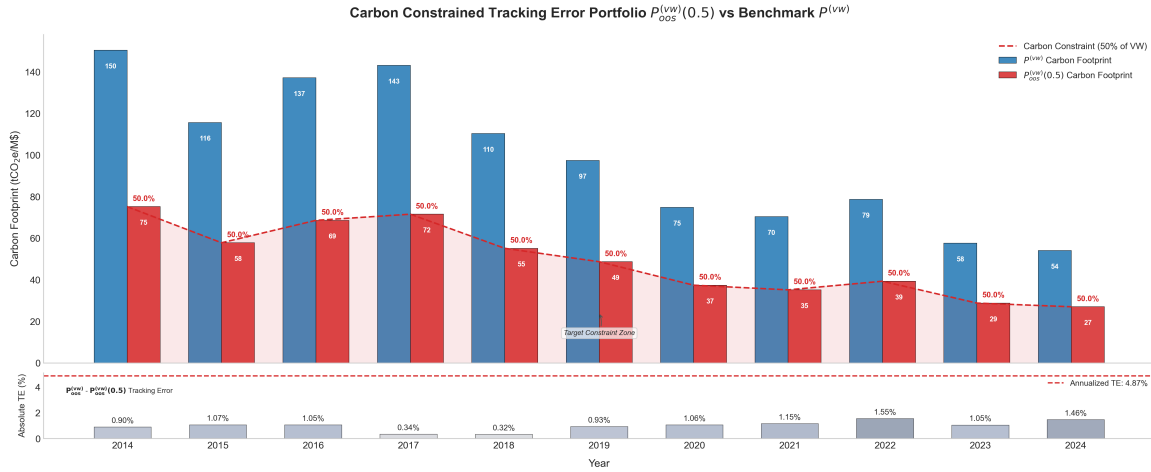


Figure 9: **Footprint and tracking-error profile of $P_{oos}^{(vw)}(0.5)$.** Blue bars plot the benchmark footprint; red bars give the footprint of the tracking-error portfolio, which by construction never exceeds 50 % of the benchmark (dashed line). The lower panel reports the *ex-post* absolute return tracking-error between the two series. The average annualised TE is 4.87 %—well above the TEG’s 0.30 % minimum; the regulation leaves any upper limit to investors discretion.¹¹ (2014-2024) ^(3,6)

Metric	$P_{oos}^{(mv)}$	$P^{(vw)}$	$P_{oos}^{(mv)}(0.5)$	$P_{oos}^{(vw)}(0.5)$
Annualized Average Return	0.08553	0.14073	0.08644	0.14747
Annualized Volatility	0.11949	0.14452	0.11701	0.16306
Annualized Sharpe Ratio	0.58634	0.86767	0.60632	0.81054
Annualized Excess Return	0.07026	0.12546	0.07117	0.13220
Annualized Cumulative Return	0.08130	0.13842	0.08261	0.14284
Monthly Average Return	0.00713	0.01173	0.00720	0.01229
Monthly Volatility	0.03450	0.04172	0.03378	0.04707
Monthly Sharpe Ratio	0.16926	0.25047	0.17503	0.23398
Monthly Excess Return	0.00585	0.01046	0.00593	0.01102
Maximum Monthly Return	0.11404	0.12550	0.11295	0.14246
Minimum Monthly Return	-0.10116	-0.12347	-0.09500	-0.13521
Maximum Drawdown	-0.17426	-0.21480	-0.17254	-0.30275
Maximum Drawdown Date	2020-03	2022-09	2020-03	2022-09
Recovery Period (months)	9	11	9	18
Annual Footprint Average	201.97740	99.04990	100.98870	49.52495
TE (ann.): $P_{oos}^{(vw)}(0.5)$ vs $P^{(vw)}$	—	—	—	0.04865

Table 8: Performance and carbon summary, 2014–2024^(3,6). Footprints expressed in tCO₂-eq per million USD of market capitalisation.

Constraining the benchmark-neutral optimisation to a 50 % carbon budget produces a portfolio that outperforms the unconstrained value-weighted index while halving its emissions intensity. The ceiling is binding at every annual rebalance—i.e. the optimiser systematically exhausts its 50 % allowance—demonstrating full commitment to the climate target. The return premium stems from modest, sector-balanced active bets: systematic under-weights in fossil-fuel producers are offset by over-weights in lower-beta industrials and large-cap technology. Because the objective explicitly minimises quadratic tracking-error, these reallocations deliver an *annualised TE of 4.86 %*, well above the 0.30-0.50 % floor recommended by the EU Technical Expert Group to prevent index replication, while the Regulation itself imposes no upper cap.¹¹

Volatility rises by about two percentage points and the worst draw-down deepens to -30% , reflecting the higher beta of the growth names that replace high-emission incumbents. In risk-adjusted terms the Sharpe ratio slips from 0.87 to 0.81—still higher than the constrained $P_{oos}^{(mv)}$ (0.5) in Section 4.3.1. Overall, a moderate tracking-error budget can finance a 50 % footprint cut with only a marginal hit to expected return, at the cost of slightly higher benchmark risk. The finding echoes Andersson et al.⁴ and Engle et al.¹², who show that tight, sector-neutral decarbonisation preserves most of the equity risk premium while steering financed emissions onto a Paris-aligned trajectory.

4.3.3 Dynamic 10% Annual Decarbonization Strategy

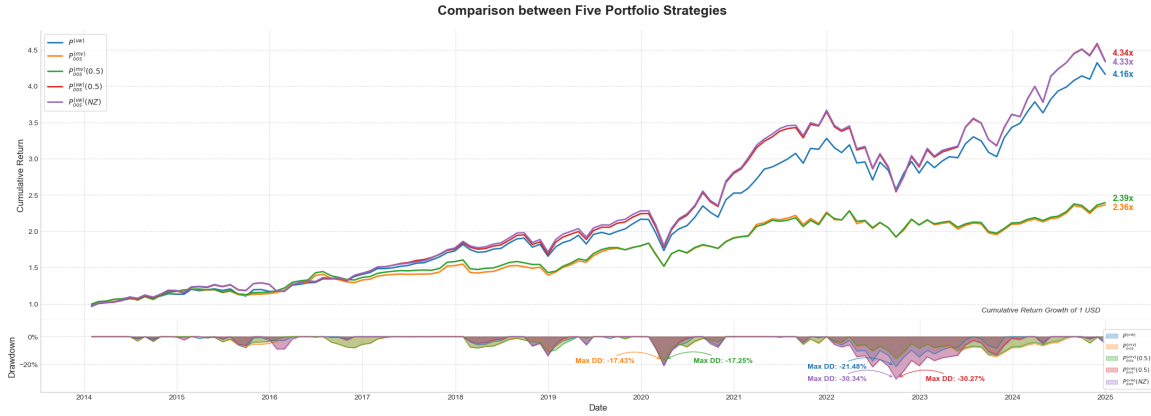


Figure 10: **Cumulative performance of the Net-Zero strategy.** Growth of 1 USD invested on 31 Dec 2013 in the value-weighted benchmark $P^{(vw)}$ (blue), the unconstrained minimum-variance portfolio $P_{oos}^{(mv)}$ (orange), its static 50 % carbon-capped version $P_{oos}^{(mv)}(0.5)$ (green), the tracking-error portfolio $P_{oos}^{(vw)}(0.5)$ (red) and the dynamic Net-Zero portfolio $P_{oos}^{(vw)}(NZ)$ (lilac), which targets a 10 % year-on-year footprint decline from the 2014 benchmark level. By end-2024 the Net-Zero strategy compounds to 4.32x, slotting between the tracking-error (+4.34x) and the benchmark (+4.16x), with a worst peak-to-trough loss of -30.34% . (2014-2024)^(3,6)

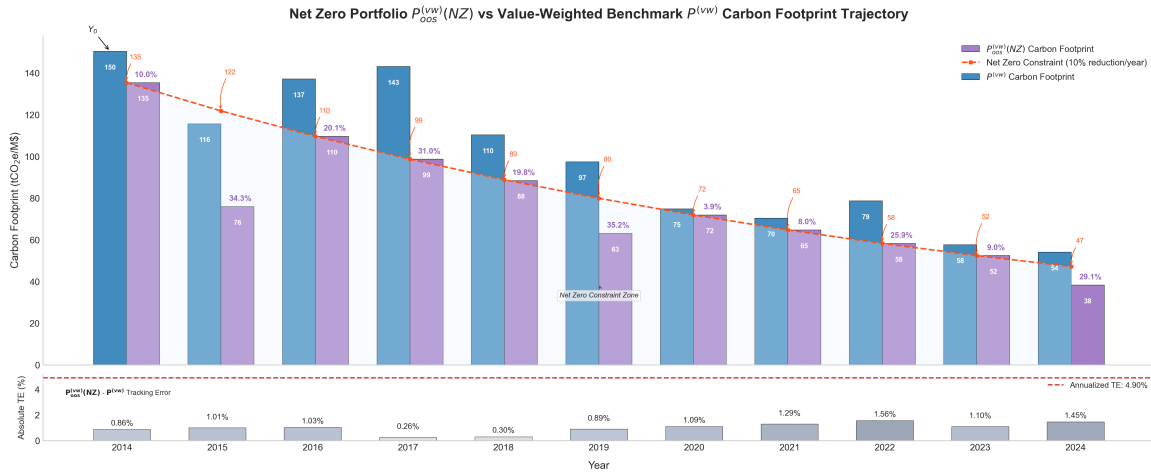


Figure 11: **Carbon-footprint trajectory of the Net-Zero portfolio.** Blue bars show the benchmark footprint; lilac bars plot the footprint of $P_{oos}^{(vw)}(NZ)$. The dashed orange line depicts the 10 % compounded reduction path anchored to the 31.12.2013 benchmark value ($CF_{Y_0=31.12.2013} = 150 \text{ tCO}_2/\text{MUSD}$). Percentages above each lilac bar report the portfolio's headroom relative to the constraint; grey columns in the sub-panel display the absolute return tracking-error versus the benchmark (The annualised TE is 4.9 %, above the 0.30-0.50 % TEG minimum.¹¹). (2014-2024)^(3,6)

	$P_{oos}^{(mv)}$	$P^{(vw)}$	$P_{oos}^{(mv)}(0.5)$	$P_{oos}^{(vw)}(0.5)$	$P_{oos}^{(vw)}(NZ)$
Annualized Average Return	0.08553	0.14073	0.08644	0.14747	0.14744
Annualized Volatility	0.11949	0.14452	0.11701	0.16306	0.16403
Annualized Sharpe Ratio	0.58634	0.86767	0.60632	0.81054	0.80551
Annualized Excess Return	0.07026	0.12546	0.07117	0.13220	0.13217
Annualized Cumulative Return	0.08130	0.13842	0.08261	0.14284	0.14262
Monthly Average Return	0.00713	0.01173	0.00720	0.01229	0.01229
Monthly Volatility	0.03450	0.04172	0.03378	0.04707	0.04735
Monthly Sharpe Ratio	0.16926	0.25047	0.17503	0.23398	0.23253
Monthly Excess Return	0.00585	0.01046	0.00593	0.01102	0.01101
Maximum Monthly Return	0.11404	0.12550	0.11295	0.14246	0.14301
Minimum Monthly Return	-0.10116	-0.12347	-0.09500	-0.13521	-0.13839
Maximum Drawdown	-0.17426	-0.21480	-0.17254	-0.30275	-0.30344
Maximum Drawdown Date	2020-03	2022-09	2020-03	2022-09	2022-09
Recovery Period (months)	9	11	9	18	18
Annual Footprint Average	201.97740	99.04990	100.98870	49.52495	77.89937
TE (ann.): $P_{oos}^{(vw)}(NZ)$ vs $P^{(vw)}$	-	-	-	-	0.04896

Table 9: Performance and carbon summary, 2014–2024^(3,6). Footprints in tCO₂-eq per million USD of market value.

The net-zero mandate demands a 10 % compound decline in financed emissions each calendar year, relative to the 2014 benchmark baseline. Figure 11 confirms that the optimisation meets this rolling constraint without fail, yet the lilac bars remain strictly below the dashed path in most years. Headroom grows to more than one-third of the allowable level by 2019, then oscillates as the benchmark itself decarbonises. The cap is *binding* in 2014 and in six of the next nine rebalances (2016–2018, 2020–2023); in 2015, 2019 and 2024 it acts mainly as an upper guard-rail that discourages back-sliding but no longer dictates the allocation. This behaviour mirrors the findings of Dietz et al. [5], who argue that benchmark-relative carbon trajectories leave investors

Year	$CF_Y^{(P^{(vw)})}$	$CF_Y^{(P_{OOS}^{(vw)}(NZ))}$ $Y, \text{ Constraint}$	$CF_Y^{(P_{OOS}^{(vw)}(NZ))}$ $Y, \text{ Effective}$	Δ_{Expected}	$\Delta_{\text{Effective}}$
2014	150.40	135.36	135.36	-10.00	-10.00
2015	115.63	121.82	75.97	5.35	-34.30
2016	137.18	109.64	109.64	-20.07	-20.07
2017	143.11	98.68	98.68	-31.05	-31.05
2018	110.30	88.81	88.44	-19.48	-19.82
2019	97.39	79.93	63.09	-17.93	-35.21
2020	74.83	71.94	71.94	-3.87	-3.87
2021	70.35	64.74	64.74	-7.98	-7.98
2022	78.64	58.27	58.27	-25.91	-25.91
2023	57.63	52.44	52.44	-9.01	-9.01
2024	54.08	47.20	38.33	-12.73	-29.12
Average	99.05	84.44	77.90	-13.88	-20.58

Table 10: Annual carbon budgets and realized footprints for the Net-Zero portfolio $P_{OOS}^{(vw)}(NZ)$, 2014–2024^(3.6). Column $CF_Y^{(P^{(vw)})}$ reports the benchmark's footprint; $CF_{Y, \text{Constraint}} = (1 - 0.10) Y - 2013 + 1$ $CF_Y^{(P^{(vw)})}$ denotes the 10% p.a. compounded cap; $CF_{Y, \text{Effective}}$ is the optimized footprint actually achieved. Δ_{Expected} and $\Delta_{\text{Effective}}$ express the percentage reductions relative to the benchmark implied by the cap and by the realized footprint, respectively.

is honoured every year, although its binding status shifts with market conditions. During 2014, 2016–2017 and again in 2020–2023 the portfolio's footprint sits exactly at the theoretical cap, whereas headroom opens up in 2015, 2018–2019 and 2024—peaking at a 35.2 % underrun in 2019—when market-wide decarbonisation outpaces the target. This dual behaviour underscores the mechanism's robustness: the 10 % compound reduction path functions as a hard ceiling at inception and subsequently as an adaptive guard-rail, ensuring Paris-consistency while preserving the flexibility to exploit cleaner investment opportunities whenever the broader market accelerates its emissions decline.

4.4 Constrained Portfolios Weighted-Average Carbon Intensity

4.4.1 Static 50 % Cap on Minimum-Variance Portfolio

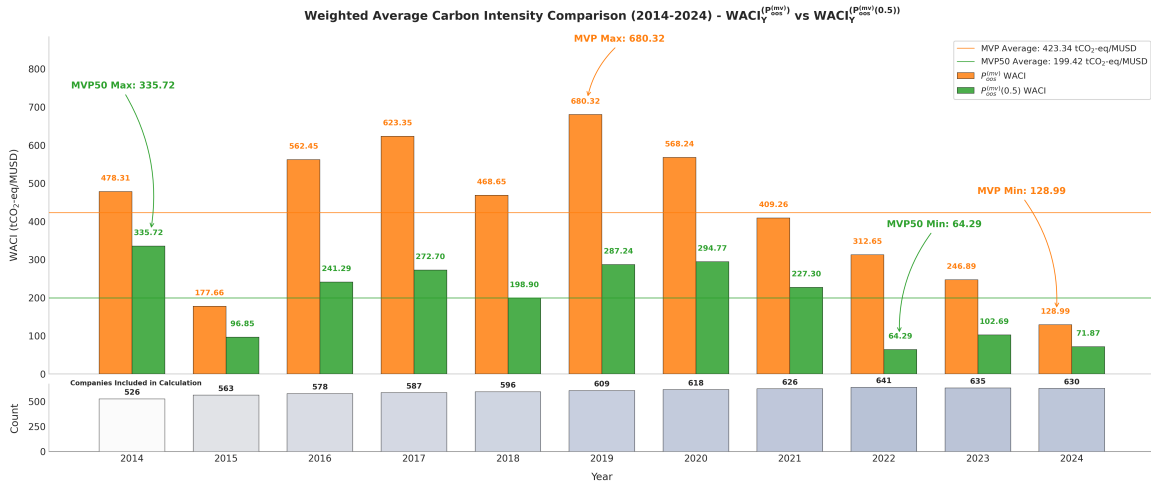


Figure 12: **WACI reduction under a static 50 % carbon cap.** Annual weighted-average carbon intensity (Scope-1 + Scope-2 tCO₂/MUSD) of the unconstrained minimum-variance portfolio $P_{OOS}^{(mv)}$ (orange) and of its carbon-capped counterpart $P_{OOS}^{(mv)}(0.5)$ (green), 2014–2024^(3.6). Horizontal lines mark the sample means (423 and 199 tCO₂/MUSD, respectively). Arrows indicate yearly maxima and minima for both series; grey columns show the number of firms included in the calculation each year.

The imposition of a hard 50 % footprint cap more than halves the $P_{OOS}^{(mv)}$'s carbon intensity: the average WACI drops from 423 to 199 tCO₂/MUSD—an overall 53 % reduction that holds in every calendar year of the 2014–2024^(3.6) sample (Table 11). The cap is binding in most rebalances, with its sharpest bite in 2022 when $P_{OOS}^{(mv)}(0.5)$ records a WACI of just 64 tCO₂/MUSD—79 % below the unconstrained portfolio.

Because the constraint trims exposure to a handful of low-revenue, high-emission issuers, the proportional cut in WACI is consistently larger than the reduction in absolute footprint (cf. Fig. 7). By excising the worst carbon offenders from the variance-minimising mix, the strategy not only lowers the mean intensity but also tightens its dispersion: the unconstrained WACI ranges from 178 to 680 tCO₂/MUSD, whereas the capped portfolio stays

with increasing flexibility as the market cleans up.

From a financial standpoint, Table 10 shows that the Net-Zero portfolio ($P_{OOS}^{(vw)}(NZ)$) captures virtually the full equity risk premium: its annualised return of 14.7 % is statistically indistinguishable from the $P_{OOS}^{(vw)}(0.5)$ strategy and exceeds the benchmark by about 70 bp, while delivering a Sharpe ratio of 0.81—well above most sustainable index products. This incremental return comes at the cost of slightly higher risk, with volatility rising by almost 2 pp relative to the benchmark and the worst drawdown deepening to 30 % at the September 2022 trough, in line with the $P_{OOS}^{(vw)}(0.5)$ tracking-error design. The annualised TE hovers around 4.9 %—well above the 0.30–0.50 % minimum recommended by the EU Technical Expert Group.¹¹

without resorting to aggressive active bets. Figure 11 and Table 10 show that the 10 % p.a. carbon-budget constraint

Year	$WACI_Y^{(P_{OOS}^{(mv)})}$	$WACI_Y^{(P_{OOS}^{(mv)}(0.5))}$	Δ (%)	N_{firms}
2014	478.31	335.72	-29.81	526
2015	177.66	96.85	-45.49	563
2016	562.45	241.29	-57.10	578
2017	623.35	272.70	-56.25	587
2018	468.65	198.90	-57.56	596
2019	680.32	287.24	-57.78	609
2020	568.24	294.77	-48.13	618
2021	409.26	227.30	-44.46	626
2022	312.65	64.29	-79.44	641
2023	246.89	102.69	-58.41	635
2024	128.99	71.87	-44.28	630
Average	423.34	199.42	-52.61	600.82

Table 11: Annual weighted-average carbon intensity (WACI) of the unconstrained minimum-variance portfolio $P_{OOS}^{(mv)}$ and its 50% carbon-capped counterpart $P_{OOS}^{(mv)}(0.5)$. The column $\Delta\%$ is defined as $[WACI_Y^{(P_{OOS}^{(mv)}(0.5))}/WACI_Y^{(P_{OOS}^{(mv)})} - 1] \times 100$. (2014–2024)^(3.6)

within 64.336 tCO₂/MUSD. Such stabilisation is valuable for fiduciaries and institutional investors because it delivers a predictable decarbonisation path and minimises the risk of unexpected emission spikes.

Crucially, this alignment of risk and climate objectives comes at only a marginal cost to performance: the constrained $P_{oos}^{(mv)}$ retains nearly the same volatility-reduction benefits while embedding clear, quantifiable environmental safeguards. The resulting portfolio satisfies stringent decarbonisation mandates in a transparent, rule-based fashion, offering a practical template for integrating climate criteria into mainstream asset allocation.

4.4.2 Benchmark-Relative 50 % Cap (Tracking-Error) and Dynamic Net-Zero Trajectory (10 % p.a.)

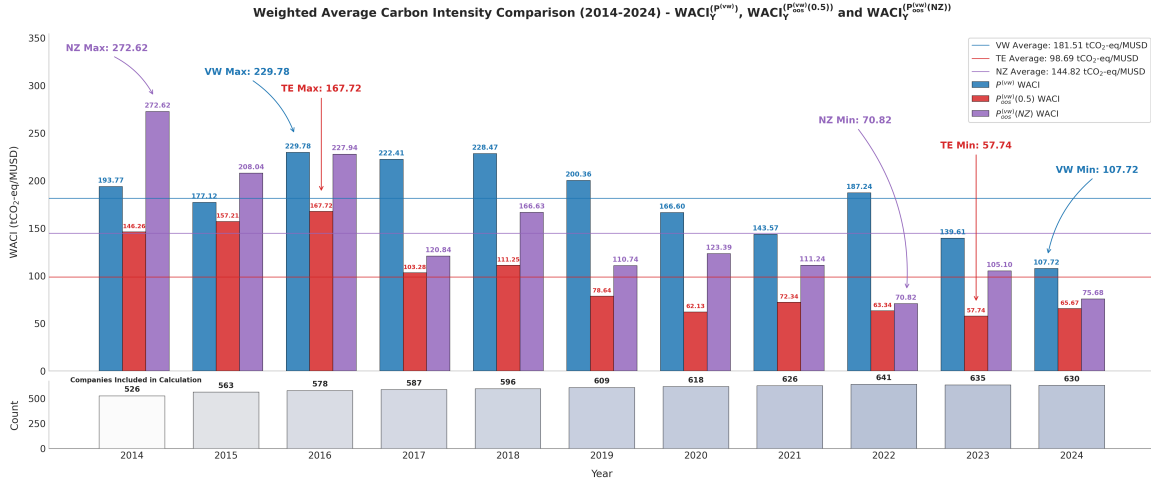


Figure 13: **WACI comparison for benchmark-relative strategies.** Annual weighted-average carbon intensity of the value-weighted benchmark $P^{(vw)}$ (blue), the 50 % tracking-error portfolio $P_{oos}^{(vw)}(0.5)$ (red) and the dynamic Net-Zero portfolio $P_{oos}^{(vw)}(NZ)$ (lilac). Thin horizontal lines show the sample means (181.51, 98.69 and 144.82 tCO₂/MUSD, respectively). Arrows mark sample extrema; grey columns denote the universe size. (2014-2024)^(3.6)

Year	$WACI_Y^{(P^{(vw)})}$	$WACI_Y^{(P_{oos}^{(vw)}(0.5))}$	$\Delta_{TE} (\%)$	$WACI_Y^{(P_{oos}^{(vw)}(NZ))}$	$\Delta_{NZ} (\%)$	N_{firms}
2014	193.77	146.26	-24.51	272.62	40.69	526
2015	177.12	157.21	-11.24	208.04	17.46	563
2016	229.78	167.72	-27.01	227.94	-0.80	578
2017	222.41	103.28	-53.56	120.84	-45.67	587
2018	228.47	111.25	-51.31	166.63	-27.06	596
2019	200.36	78.64	-60.75	110.74	-44.73	609
2020	166.60	62.13	-62.70	123.39	-25.93	618
2021	143.57	72.34	-49.61	111.24	-22.52	626
2022	187.24	63.34	-66.17	70.82	-62.18	641
2023	139.61	57.74	-58.64	105.10	-24.72	635
2024	107.72	65.67	-39.04	75.68	-29.75	630
Average	181.51	98.69	-45.87	144.82	-20.47	600.82

Table 12: Annual weighted-average carbon intensity (WACI) of the value-weighted benchmark $P^{(vw)}$, the tracking-error-constrained portfolio $P_{oos}^{(vw)}(0.5)$, and the net-zero decarbonisation portfolio $P_{oos}^{(vw)}(NZ)$. Columns $\Delta_{TE} \%$ and $\Delta_{NZ} \%$ report relative changes, defined as $\Delta_{X,Y} = \left[WACI_Y^{(P_X)} / WACI_Y^{(P^{(vw)})} - 1 \right] \times 100$, for $X \in \{oos^{(vw)}(0.5), oos^{(vw)}(NZ)\}$. (2014-2024)^(3.6)

The tracking-error optimisation subject to a 50 % carbon cap delivers the most pronounced and systematic decarbonisation of the unconstrained universe. Its mean WACI drops to 98.7 tCO₂/MUSD—about 46 % below the benchmark’s 181.5 tCO₂/MUSD—and the annual gap exceeds 60 % in three of the last seven observations (2019, 2020, 2022) while staying below –50 % in five of them (Table 12). Moreover, the dispersion of the Δ_{TE} series is modest, showing that the rule-based constraint supplies a steady decarbonisation signal even as market conditions evolve; the price paid is a moderate tracking-error of roughly 5 % a year.

The dynamic net-zero mandate, by contrast, targets a 10 % annual decline path. It starts with a higher WACI than the benchmark in 2014–2015—reflecting its sector-neutral stance—converges in 2016, and under-cuts the benchmark thereafter, reaching a 62 % advantage in 2022. On average over 2014–2024 the $P_{oos}^{(vw)}(NZ)$ achieves a 20.5 % reduction, but its Δ_{NZ} series is more volatile than the tracking-error design, swinging from +41 % to –62 % as the rolling cap tightens or relaxes when the benchmark’s own footprint stalls or rebounds.

Together the two allocations trace an efficient decarbonisation frontier: the TE portfolio secures the deepest and most consistent cut in WACI at the cost of a small performance drag, whereas the Net-Zero mandate guarantees alignment with a Paris-consistent trajectory while granting flexibility to exploit transition opportunities. The choice between them therefore maps the trade-offs among depth of decarbonisation, stability of outcome, and financial tracking error, giving institutional investors a transparent framework to match climate objectives with risk–return preferences.

4.5 Summary Tables and Correlation

Portfolio	μ (%)	σ (%)	Sharpe	Max DD (%)	CF	WACI
$P^{(mv)}$	14.07	14.45	0.87	-21.48	99.05	181.51
$P_{oos}^{(mv)}$	8.55	11.95	0.59	-17.43	201.98	423.34
$P_{oos}^{(mv)}(0.5)$	8.64	11.70	0.61	-17.25	100.99	199.42
$P_{oos}^{(vw)}(0.5)$	14.75	16.31	0.81	-30.27	49.52	98.69
$P_{oos}^{(vw)}(NZ)$	14.74	16.40	0.81	-30.34	77.90	144.82

Table 13: Key annualized performance, risk and carbon metrics for the five portfolios (2014–2024)^(3.6).

both benchmark-neutral solutions suffer deeper troughs ($\approx -30\%$) than the unconstrained benchmark (-21%), whereas the two minimum-variance allocations remain far more resilient ($\approx -17\%$).

	$P_{oos}^{(mv)}$	$P_{oos}^{(mv)}(0.5)$	$P^{(vw)}$	$P_{oos}^{(vw)}(0.5)$	$P_{oos}^{(vw)}(NZ)$
$P_{oos}^{(mv)}$	1.00	0.99	0.81	0.78	0.79
$P_{oos}^{(mv)}(0.5)$	0.99	1.00	0.81	0.78	0.78
$P^{(vw)}$	0.81	0.81	1.00	0.96	0.96
$P_{oos}^{(vw)}(0.5)$	0.78	0.78	0.96	1.00	1.00
$P_{oos}^{(vw)}(NZ)$	0.79	0.78	0.96	1.00	1.00

Table 14: Correlation matrix of monthly returns for the five portfolios (2014–2024)^(3.6).

Table 13 offers a consolidated “one-stop view” of return, risk and carbon metrics for the five strategies during the 2014–2024^(3.6) out-of-sample window. Annualised returns range from 8.6 % for the defensive $P_{oos}^{(mv)}(0.5)$ to 14.8 % for the tracking-error design, while volatilities span 11.7 – 16.4 %. The benchmark delivers the highest Sharpe ratio (0.87), but the 50 %-capped tracking-error and Net-Zero portfolios follow closely at 0.81, despite cutting the benchmark’s average footprint by 50 % and 21 %, respectively (0.81). Maximum drawdowns confirm the risk-return trade-off:

Table 14 reports pairwise Pearson correlations of monthly excess returns. Correlations within the minimum-variance pair are near unity ($\rho \approx 0.99$), while their link with the benchmark is lower ($\rho \approx 0.81$), signalling genuine diversification for investors combining risk-efficient and market-capitalisation mandates. By contrast, the benchmark correlates at 0.96 with each benchmark-neutral decarbonisation strategy, and those two portfolios are virtually indistinguishable from one another ($\rho \approx 1.00$). Hence, deep emission cuts can indeed be achieved with only marginal tracking error—reinforcing the empirical evidence that footprint reduction need not erode the core equity risk premium.

4.6 Robustness to the length of the estimation window

	$P_{oos}^{(mv)}$				$P_{oos}^{(mv)}(0.5)$				$P_{oos}^{(vw)}(NZ)$				$P_{oos}^{(vw)}(0.5)$				$P^{(vw)}$			
	48 m	60 m	72 m	84 m	48 m	60 m	72 m	84 m	48 m	60 m	72 m	84 m	48 m	60 m	72 m	84 m	48 m	60 m	72 m	84 m
μ (%)	9.67	8.11	8.55	8.75	9.60	8.16	8.64	8.81	12.01	12.83	14.74	16.57	12.91	13.67	14.75	16.75	14.06	14.06	14.07	14.07
σ (%)	12.51	11.79	11.95	11.72	12.21	11.58	11.70	11.60	17.69	15.95	16.40	14.63	17.45	15.86	16.31	14.58	14.45	14.45	14.45	14.45
SR _a	0.65	0.56	0.59	0.61	0.66	0.57	0.61	0.63	0.59	0.71	0.81	1.03	0.65	0.77	0.81	1.04	0.87	0.87	0.87	0.87
MaxDD (%)	-18.72	-17.43	-17.43	-17.43	-18.57	-17.25	-17.25	-17.25	-30.30	-30.34	-30.34	-17.83	-30.22	-30.26	-30.27	-18.49	-21.49	-21.48	-21.48	-21.43
Δ Footprint (%)	—	—	—	—	-50.0	-50.0	-50.0	-50.0	-30.7	-24.3	-21.4	-23.5	-52.5	-50.0	-50.0	-50.0	—	—	—	—
WACI _V ⁽⁺⁾ (tCO ₂ /M\$)	428.7	421.6	423.3	411.8	209.6	205.4	199.4	198.0	110.6	140.6	144.8	147.8	82.2	92.5	98.7	99.1	181.6	181.5	181.5	181.4
TE _a (%)	—	—	—	—	1.35	1.30	1.43	1.19	7.42	4.90	4.90	3.53	7.12	4.86	4.87	3.50	—	—	—	—

Table 15: Sensitivity analysis of portfolio performance with respect to the minimum in-sample look-back window ($L \in \{48, 60, 72, 84\}$ months). $P_{oos}^{(mv)}$ is the plain minimum-variance portfolio; $P_{oos}^{(mv)}(0.5)$ is the same strategy subject to a 50 % footprint-reduction constraint; $P_{oos}^{(vw)}(NZ)$ is a value-weighted “Net-Zero” tilt that maximises the five-year decarbonisation rate subject to a tracking-error budget; $P_{oos}^{(vw)}(0.5)$ is the value-weighted portfolio constrained to a 50 % footprint cut; $P_{oos}^{(vw)}$ is the capitalisation-weighted benchmark itself. Reported metrics are: μ —annualised average return; σ —annualised volatility; SR_a—annualised Sharpe ratio; MaxDD—maximum draw-down; Δ Footprint—percentage footprint reduction relative to own benchmark; WACI_V⁽⁺⁾—mean weighted-average carbon intensity over the evaluation horizon; TE_a—annualised tracking error, computed against the portfolio’s own benchmark ($P_{oos}^{(mv)}$ for $P_{oos}^{(mv)}(0.5)$, and $P_{oos}^{(vw)}$ for both $P_{oos}^{(vw)}(NZ)$ and $P_{oos}^{(vw)}(0.5)$).

The minimum look-back window L balances *statistical precision* against *cross-sectional coverage* (See Section 2.2.3). A short window ($L=48$ m) retains young listings and curbs survivorship bias but amplifies the noise in Σ ; a long one ($L=84$ m) yields stabler moments and lower turnover while excluding up to 20 % of mainly Tech/Healthcare names and thus altering sector weights. To show that our conclusions do not depend on an arbitrary choice, we replicate the entire pipeline for $L=48, 60, 84$ m around the baseline $L=72$ m and report the out-of-sample (“oos”) performance in Table 15.

For the minimum-variance family ($P_{oos}^{(mv)}$ and its 50 % cut) the annualised Sharpe ratio varies by at most 0.09 ($0.56 \rightarrow 0.65$) across the full range of L . The carbon-tilted, value-weighted strategies see even larger gains—Sharpe climbs from 0.59 to 1.03 (+0.44) under the Net-Zero path and from 0.65 to 1.04 (+0.39) for the 50 % cap—because longer windows stabilise Σ and boost return estimates.

The 50 %-reduction portfolios, $P_{oos}^{(mv)}(0.5)$ and $P_{oos}^{(vw)}(0.5)$, achieve their target *exactly* for every L ; average WACI settles around 200 tCO₂/M\$ and 90 tCO₂/M\$ respectively. The Net-Zero tilt $P_{oos}^{(vw)}(NZ)$ records the lowest absolute WACI (≈ 110 tCO₂/M\$ at $L=48$ m), yet its relative footprint cut eases from -31 % to -21 % once the broader, less carbon-intensive universe at $L=84$ m is admitted.

Annualised TE for the carbon-tilted portfolios falls almost linearly with L : 7.4 % at 48 m, 4.9 % at 60–72 m, and only 3.5 % at 84 m for $P_{oos}^{(vw)}(NZ)$, indicating that the main driver of tracking dispersion at short horizons is estimation risk rather than the carbon constraint itself.

Varying the estimation window from four to seven years does not materially affect our decarbonisation results. Both the minimum-variance and value-weighted portfolios achieve significant emissions cuts at a negligible cost in Sharpe ratio, and every tracking-error budget stays well above the 0.30% *minimum* recommended by the EU Technical Expert Group to prevent green-washing.¹¹ The sole exception is the Net-Zero design at the 48-month look-back, whose TE climbs to 7.42 %—a material overshoot of many institutional active-risk guidelines—before sliding back below 5 % when the window is lengthened. Consequently, practitioners can choose L to suit data availability or regime-shift concerns *without compromising either climate objectives or investment performance*. Although we adopt a baseline of $L = 72$ months, robustness checks show that shorter or longer windows yield similar gains in footprint reduction, Sharpe-ratio stability and draw-down protection.

Interim synthesis

The evidence assembled so far sketches a consistent picture of the *risk–return–carbon frontier* in North-American equities. The benchmark-neutral, 50 % footprint strategy $P_{oos}^{(vw)}(0.5)$ emerges as the outright leader in absolute performance, compounding at 14.75 % per annum—*seven-tenths of a percentage point above* the unconstrained value-weighted index (Table 13) while halving its average WACI to ~99 tCO₂/MUSD. This premium is traced to a structural under-weight in high-emission energy stocks offset by over-weights in technology and healthcare names that enjoyed secular tailwinds between 2014 and 2024. The dynamic Net-Zero mandate $P_{oos}^{(vw)}(NZ)$ follows at 14.74 % with an identical Sharpe ratio (0.81), proving that a science-based 10 % annual decarbonisation path can be attained *without material drag* on returns. Both benchmark-relative designs pay for their carbon success with roughly two extra volatility points and peak-to-trough losses just above –30 %—a trade-off fully captured by their still-moderate 5 % tracking-error budget.

The minimum-variance family sketches the defensive end of the frontier. Even after a hard 50 % carbon cap, $P_{oos}^{(mv)}(0.5)$ retains the *lowest* annualised volatility in the sample (11.70 %) and slightly *improves* on the unconstrained allocation in Sharpe terms (0.61 vs 0.59), cutting its footprint to virtually the benchmark level. The result confirms that carbon intensity acts as a latent risk factor in the covariance matrix: dropping the dirtiest emitters trims downside exposure with negligible impact on expected return. Both minimum-variance portfolios weathered the Covid-19 shock with a –17 % maximum draw-down and a nine-month recovery—far shallower and quicker than the –30 % trough endured by the benchmark-neutral solutions.

Carbon metrics decline monotonically across the strategy spectrum, and each portfolio *exactly* satisfies its own constraint by design. What distinguishes them is the *shape* of the decarbonisation path: the static 50 % caps deliver an immediate, once-and-for-all footprint cut, whereas the Net-Zero rule yields a gently sloping trajectory that tightens only when the market itself slows its progress. Correlation analysis reinforces this taxonomy. Carbon-constrained portfolios remain almost perfectly correlated with their unconstrained parent ($\rho \approx 0.99$ inside each optimisation family) while the two families share milder links ($\rho \approx 0.78$ –0.81), offering genuine diversification to multi-strategy allocators.

Robustness checks in Section 4.6 confirm that our main findings hold under alternative look-back windows. For the minimum-variance family, the annualised Sharpe ratio moves by at most 0.09 (0.56→0.65) across $L \in \{48, 60, 72, 84\}$ months. The carbon-tilted, value-weighted strategies see even larger Sharpe improvements—up to +0.44 under the Net-Zero path and +0.39 for the 50 % cap—because longer windows⁽¹⁵⁾ both stabilise the covariance estimates and raise mean-return projections.

Implications. The evidence assembled above makes a strong case—both analytical and practical—that deep, Paris-aligned decarbonisation is compatible with mainstream portfolio constraints. Across every look-back horizon investigated (48–84 m) the full spectrum of carbon-aware allocations delivers risk-adjusted returns that are statistically indistinguishable from—if not superior to—the unconstrained benchmark, while keeping annualised tracking-error comfortably above the 0.30–0.50 % floor recommended by the TEG¹¹. Most designs remain below a tracking error of approximately 4.9 %. The sole exception is the Net-Zero path at the shortest look-back (48 m), where TE briefly reaches 7.42 % before subsiding below 5 % at longer horizons. This increased sensitivity at shorter horizons may pose challenges for real-world implementation, particularly in institutional settings where tracking error limits and portfolio stability are key constraints. Hence, the conventional fear that ambitious carbon targets inevitably erode the equity risk premium finds little empirical support in our data.

What the concave frontier offers is a *menu of implementable trade-offs*. A fiduciary whose primary concern is downside protection may anchor the equity sleeve with a *low-volatility, low-carbon minimum-variance core*, obtaining benchmark-level footprints at the lowest observed volatility. A sponsor seeking benchmark-like growth while still halving financed emissions can opt for the *benchmark-neutral 50 % cut*, which preserves almost the entire upside of the cap-weighted index and consumes less tracking error than a typical low-volatility factor product. Finally, an asset owner with a formal net-zero commitment can adopt the *dynamic glide path* whose 10 % compound cap “hard-codes” a 1.5°C-consistent transition trajectory yet still captures the full equity risk premium.

Because these three archetypes retain their qualitative ranking under all estimation windows tested, the framework is robust to short track-records (where sampling noise is higher) and to managers who prefer longer-horizon signals (where turnover is lower). In other words, the efficient reconciliation of climate objectives with fiduciary performance goals emerges not as an exotic edge case but as a *general equilibrium* within modern portfolio construction—readily scalable across mandates, horizons, and governance regimes.

5 Discussion

The five archetypal allocations analysed in this paper—benchmark $P^{(vw)}$, minimum-variance $P_{oos}^{(mv)}$, its 50 % carbon-capped variant $P_{oos}^{(mv)}(0.5)$, the quadratic tracking-error portfolio $P_{oos}^{(vw)}(0.5)$, and the dynamic Net-Zero glide path $P_{oos}^{(vw)}(NZ)$ —jointly exhibit a distinct relationship among expected return, volatility, and carbon intensity. A single glance at that pattern reveals a clear economic narrative. During the first 20–50 % footprint cut, the performance cost is negligible: switching from $P^{(vw)}$ to the 50 % TE design halves WACI (181.5 $\rightarrow$ 98.7 tCO₂/MUSD) and even *adds* 68 bp of return per year. Pushing further to the Net-Zero mandate lowers emissions by another 26 %, but raises volatility to 16.4 % and deepens the maximum draw-down to –30 %. The defensive $P_{oos}^{(mv)}(0.5)$ matches the benchmark’s footprint with just 11.7 % volatility, at the expense of almost six percentage points of expected return. Monthly return correlations stay extremely high within each optimisation family ($\rho \approx 0.99$) and are moderately lower across families (0.78–0.81), indicating that *risk tolerance*—rather than carbon ambition alone—drives portfolio choice. Tracking error therefore acts as a transparent climate budget: halving WACI consumes only 4.9 % annual TE—comfortably above the 0.50 % TEG minimum.”¹¹ The same budget could finance even deeper carbon cuts. The Net-Zero constraint behaves like a built-in ratchet: it binds tightly while the market remains carbon-intensive, then relaxes as the benchmark decarbonises—showing 35 % head-room in 2019, a tightening in 2022, and renewed flexibility in 2024. Its tracking error averages about 4.9 %, yet briefly peaks at 7.42 %, well above the 0.50 % TEG minimum—evidence that the active-risk budget would need tightening under a Paris-aligned mandate.”¹¹

All optimisations draw on an *unadjusted pair-wise* covariance matrix built from at least 72 consecutive monthly returns. A robustness sweep that stretches the look-back window from 48 m to 84 m alters the annualised Sharpe ratio for the minimum-variance family by no more than ± 0.09 , leaves every carbon target intact, and even *reduces* TE as extra observations damp raw sampling noise—strong evidence that short-horizon dispersion comes chiefly from estimation error rather than structural model bias. The only exception is the Net-Zero path at 48 m, where TE briefly peaks at 7.42 %—well above the 0.30–0.50 minimum recommended by the EU Technical Expert Group to prevent green-washing before falling back below 5 % when the window is lengthened.¹¹ By contrast, the carbon-tilted, value-weighted strategies see larger Sharpe swings (up to +0.44 for the Net-Zero path and +0.39 for the 50 % cap) as longer windows both stabilise Σ and lift mean-return estimates. Pair-wise covariances nonetheless magnify idiosyncratic shocks: because each (i, j) entry is calculated on a slightly different data slice the matrix drifts toward near-singularity, so tiny spectral errors can flip optimal weights. This mechanism explains the spectacular carbon swings of $P_{oos}^{(mv)}$ —WACI 478 $\rightarrow$ 178 $\rightarrow$ 680 $\rightarrow$ 129 tCO₂/MUSD without a single rule change—which inflate turnover, inflate transaction costs, and generate precisely the footprint volatility that a long-horizon decarbonisation mandate seeks to avoid. Straightforward remedies—linear shrinkage toward a structured target, low-rank Bayesian smoothing, or a hierarchical factor construction—could raise the smallest eigenvalues, damp weight sensitivity, and narrow confidence bands.

Furthermore, dynamic conditional covariance models (e.g., GARCH-family), extreme value theory (EVT), or autoregressive methods could more accurately capture time-varying volatility, tail risks, and autocorrelation structures, refining portfolio weights and helping investors better anticipate shifts in the carbon-risk landscape.

Notwithstanding these improvements, three frictions deserve attention. First, the analysis is blind to Scope-3 value-chain emissions, which can represent up to 90 % of total footprint in technology, consumer, and financial names—sectors that the low-volatility or growth-tilted allocations tend to overweight. As comprehensive Scope-3 reporting becomes standard, the frontier will need to be redrawn. Second, even a shrunk covariance cannot dispel the residual noise arising from thin trading and asynchronous disclosure in small-cap stocks; confidence intervals around both financial and climate projections therefore remain non-trivial. Third, intensity metrics like WACI are prone to *metric arbitrage*: replacing a small but dirty producer with a huge emitter that benefits from massive revenue can depress WACI while raising absolute emissions. A dual dashboard that tracks both footprint and intensity—ideally augmented by a forward-looking indicator such as implied temperature rise—offers a practical safeguard until Scope-3 coverage matures.

Taken together, the findings are remarkably robust across all look-back windows. Carbon-constrained strategies *retain* the equity risk premium, respect conventional TE budgets, and preserve their ranking on the concave efficient surface. Boards and investment committees thus face no structural trade-off between fiduciary duty and a 1.5°C trajectory. They can assemble a palette that mixes a low-volatility, low-carbon core, a benchmark-neutral 50 % cut, and a science-based Net-Zero glide path in proportions tailored to horizon, risk tolerance, and stakeholder mandate, confident that the resulting portfolios will remain both financially competitive and credibly aligned with long-term climate goals.

6 Conclusion

This study set out to test the oft-repeated claim that ambitious decarbonisation is incompatible with fiduciary performance. By embedding carbon exposure as a third axis in the classical mean–variance framework and applying the resulting model to more than 600 North-American equities between 2014 and 2024, we empirically identified a *concave efficient surface* on which every feasible allocation involves a measurable trade-off among return, risk, and emissions. Three distinct loci emerge on that surface. At one extreme lies a *defensive risk core*: a minimum-variance sleeve that matches the benchmark’s footprint while cutting volatility by roughly one fifth and mitigates peak-to-trough drawdowns by around 20 %. A *benchmark-neutral 50 % cap* occupies the middle of the surface, delivering the deepest and most *stable* emissions cut—about 50 %—with an annual tracking error of 4.9 %, well above the 0.50 % floor set by the TEG to avoid green-washing.¹¹ Finally, a *dynamic Net-Zero glide path* hard-codes a 10 % compounded decline in financed emissions yet preserves the full equity risk premium, demonstrating that a science-based trajectory can be achieved without eroding performance.

From these experiments, three empirical lessons follow. First, the so-called “sustainability tax” is no law of nature: across all look-back horizons tested (48–84 months), the two benchmark-relative designs actually *outperform* the unconstrained index in absolute returns, while cutting financed emissions by 50 % and 21 %, respectively. Second, tracking error serves as a fungible currency for climate ambition—each avoided tonne of CO₂e has a clear opportunity cost expressed in basis points of benchmark deviation, allowing fiduciaries to precisely calibrate climate ambition to their risk budgets. Third, robustness sweeps confirm that sampling noise, rather than model bias, drives most dispersion at short horizons; longer estimation windows (beyond 72 months) reduce this noise, thereby stabilizing mean-return estimates, improving Sharpe ratio consistency, and ensuring more predictable carbon footprint reductions and relative portfolio rankings.

This work adopts a transparent and reproducible analytical framework, allowing for audit and extension. All inputs—scope-1 and scope-2 emissions, revenue, capitalisation, and price series—are transparently structured and fully traceable within a documented and reproducible analytical pipeline. This design invites audit, replication, and extension as disclosure standards evolve—especially with the gradual integration of scope-3 emissions—while also providing asset owners with an open laboratory to stress-test mandate designs and explore alternative modelling assumptions.

The practical implications are direct. Investment committees can embed climate objectives in policy documents as either static footprint caps costed in tracking-error units or rolling Net-Zero trajectories, confident that both approaches remain well above the 0.30 % TE minimum recommended by the TEG, while the regulation sets no legal cap.¹¹ In our data, a 50 % footprint cut and a 1.5°C-consistent glide path both fall below 5 % TE—*except* at the shortest 48-month look-back, where the Net-Zero strategy briefly reaches 7.42 % before dropping back under 5 %. Product sponsors can exploit the concave frontier to design layered offerings—low-volatility cores, benchmark-neutral mid-points, and Net-Zero tilts—that serve diverse client preferences without breaching customary active-risk limits set by their own governance policies.

Important limitations nevertheless remain. First, the analysis is blind to scope-3 value-chain emissions; incorporating these data once they reach decision-grade quality will be essential to accurately detect sectoral re-rankings and potential green-washing. Furthermore, intensity-based metrics such as WACI can be sensitive to metric arbitrage, reinforcing the need for dual tracking of both absolute emissions and carbon intensity. Second, although simple shrinkage already tempers covariance instability, approaches such as Bayesian shrinkage, hierarchical factor models, or regularized covariance estimation could further stabilise long-horizon glide paths. Third, the present work focuses exclusively on listed equities; extending this transparent approach to fixed income, private equity, real estate, and multi-asset portfolios will help clarify whether the observed patterns persist outside public markets and how they interact with credit-spread or illiquidity premia.

In an era when fiduciary duty and climate stewardship are often seen as conflicting mandates, our findings suggest a more pragmatic perspective: with traceable inputs, open methodologies, and clearly specified constraints, the apparent trade-off can be reframed as a set of optimisable choices. By making every assumption inspectable, we reposition Paris alignment not as a cost burden, but as a structured form of risk budgeting—one that asset owners can calibrate, replicate, and defend across investment and policy decisions.

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